

ON PROPERTIES OF KAC-MOODY LIE ALGEBRAS IN DELIGNE CATEGORIES

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ABSTRACT. We generalize the notion of Kac-Moody Lie algebras to Deligne Categories. We then construct some important features of these Lie algebras in the abstract setting. Finally, using our construction of the Kac-Moody Lie algebra, we define and prove some properties of the Bernstein-Gelfand-Gelfand category $\mathcal{O}$ as well as an abstract Shapovalov form on highest weight Verma modules in this category.

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1. INTRODUCTION

The original work [4] of P. Deligne and J.S. Milne treats the theory of Tannakian categories, i.e. tensor categories over a field $\mathbb{k}$ admitting a strong monoidal functor (called fiber functor) to the category of vector spaces $Vec_{\mathbb{k}}$ over $\mathbb{k}$. As an example of non-Tannakian categories, they give examples of categories corresponding classical groups GL_n, O_n, Sp_{2n} which generalize to arbitrary parameter n and behave polynomially in it. Later P. Deligne gives an example of S_n (in [3]). Those categories fail to admit a fiber functor since objects from them have a non-integer categorical dimension. Furthermore, Deligne's theorem [2] states that a symmetric tensor category over a field $\mathbb{k}$ of characteristic zero is Tannakian if and only if tensor power dimensions satisfy specific moderate growth bounds. It can be shown that all the categories above fail to satisfy the growth condition.

In [6, 7] P. Etingof has generalized many usual representation theoretic objects in the setting of the above-mentioned tensor categories for $n \in \mathbb{C}$. Some of the more recent results include generalization of the Schur-Weyl duality [5], the Weyl-Kac character formula for affine Lie algebras [15], Yangians [12], Harish-Chandra bimodules [17], Feigin–Frenkel duality [16] etc. Also, a description of the category $\mathcal{O}$ of modules over $\mathfrak{gl}_{\mathbf{n}}$ in the case of $\boxtimes_{i=1}^m \underline{\text{Rep}}(\mathfrak{gl}_{\mathbf{n}_i})$ with $\mathbf{n} = \sum_{i=1}^m \mathbf{n}_i$ has been done in [18].

1.1. Structure of the paper. We begin by reviewing the preliminary concepts of Kac-Moody Lie algebras, tensor categories, and Deligne categories in Section 2. In Section 3, we provide a construction of categorical Kac-Moody Lie algebras in a Deligne category setting. In Section 4, we generalize some of the properties of classical Kac-Moody Lie algebras to our construction. Finally, in Section 5, we generalize the notions of the Bernstein-Gelfand-Gelfand category $\mathcal{O}$, Verma modules, and the contravariant Shapovalov form.

2. BACKGROUND

2.1. Kac-Moody Lie Algebras. In this section, we briefly recall the definition and basic properties of Kac-Moody Lie algebras from [11], which generalize the theory of finite-dimensional semisimple Lie algebras. Let $A = (a_{ij})_{i,j=1}^n$ be a generalized Cartan matrix, which satisfies $a_{ii} = 2$, $a_{ij} \in \mathbb{Z}_{\leq 0}$ for $i \neq j$, and $a_{ij} = 0$ if and only if $a_{ji} = 0$.

A *realization* of A is a triple $(\mathfrak{h}, \Pi, \Pi^\vee)$ where

- $\mathfrak{h}$ is a complex vector space with dimension $2n - \text{rank}(A)$,
- $\Pi = \{\alpha_1, \dots, \alpha_n\} \subset \mathfrak{h}^*$ is a set of linearly independent simple roots, and
- $\Pi^\vee = \{\alpha_1^\vee, \dots, \alpha_n^\vee\} \subset \mathfrak{h}$ is a set of linearly independent simple coroots,

satisfying $\alpha_j(\alpha_i^\vee) = a_{ij}$ for all i and j .

The *Kac-Moody Lie algebra* $\mathfrak{g}(A)$ associated to the matrix A is the Lie algebra generated by the Cartan subalgebra $\mathfrak{h}$ and the Chevalley generators e_i, f_i ($1 \leq i \leq n$), subject to the following relations:

$$\begin{aligned} [h, h'] &= 0 \quad \text{for all } h, h' \in \mathfrak{h}, \\ [h, e_i] &= \alpha_i(h)e_i, \\ [h, f_i] &= -\alpha_i(h)f_i, \\ [e_i, f_j] &= \delta_{ij}\alpha_i^\vee, \\ (\text{ad } e_i)^{1-a_{ij}}(e_j) &= 0 \quad \text{for } i \neq j, \\ (\text{ad } f_i)^{1-a_{ij}}(f_j) &= 0 \quad \text{for } i \neq j. \end{aligned}$$

The last two relations are known as the *Serre relations*. $\mathfrak{g}(A)$ has a *root space decomposition* $\mathfrak{g}(A) = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha$, where $\mathfrak{g}_\alpha = \{x \in \mathfrak{g} \mid [h, x] = \alpha(h)x \text{ for all } h \in \mathfrak{h}\}$ (i.e., the joint eigenspace decomposition with respect to $\mathfrak{h}$). If $\mathfrak{g}_\alpha \neq 0$, then α is a *root*, and the set of roots Δ is known as the *root system*. Define $Q = \sum \mathbb{Z}\alpha_i$ to be the integer lattice generated by the simple roots, $Q_+ = \sum \mathbb{Z}_{\geq 0}\alpha_i$, and $\Delta_+ = \Delta \cap (Q_+ \setminus \{0\})$. Roots in Δ_+ are called *positive roots*.

The Kac-Moody algebra $\mathfrak{g}(A)$ admits a canonical involutive automorphism $\omega : \mathfrak{g}(A) \rightarrow \mathfrak{g}(A)$ known as the *Cartan involution*. It is uniquely defined on the generators by:

$$(1) \quad \omega(e_i) = -f_i, \quad \omega(f_i) = -e_i, \quad \omega(h) = -h \quad \text{for } h \in \mathfrak{h}.$$

The Cartan involution exchanges the positive and negative Borel subalgebras, mapping the root space $\mathfrak{g}_\alpha$ to $\mathfrak{g}_{-\alpha}$.

A generalized Cartan matrix A is said to be *symmetrizable* if there exists an invertible diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$ and a symmetric matrix B such that $A = DB$.

Theorem 2.1 ([11]). *Let $\mathfrak{g}(A)$ be a Kac-Moody algebra associated to a symmetrizable matrix A . Fix a decomposition of A (as $A = DB$). Then there exists a non-degenerate symmetric bilinear $\mathbb{C}$ -valued form $(\cdot|\cdot)$ on $\mathfrak{g}(A)$ such that:*

- a) $(\cdot|\cdot)$ is invariant, i.e., $([x, y]|z) = (x|[y, z])$ for all $x, y, z \in \mathfrak{g}(A)$.
- b) $(\cdot|\cdot)|_{\mathfrak{h}}$ is non-degenerate.
- c) $(\mathfrak{g}_\alpha|\mathfrak{g}_\beta) = 0$ if $\alpha + \beta \neq 0$.
- d) $(\cdot|\cdot)|_{\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}}$ is non-degenerate for $\alpha \neq 0$, and hence $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ are non-degenerately paired by $(\cdot|\cdot)$.

On the Cartan subalgebra $\mathfrak{h}$, this form is determined by the condition $(\alpha_i^\vee|h) = d_i \alpha_i(h)$ for all $h \in \mathfrak{h}$. This naturally induces a non-degenerate symmetric bilinear form on $\mathfrak{h}^*$ and an isomorphism $\nu: \mathfrak{h} \xrightarrow{\sim} \mathfrak{h}^*$ given by $\langle \nu(h), h' \rangle = (h|h')$.

2.2. Tensor Categories and Lie Algebra Objects. To generalize the classical notion of Lie algebras, we move to the setting of symmetric tensor categories. Let $\mathcal{C}$ be a *symmetric tensor category* over a field $\mathbb{k}$ of characteristic zero. By this, we mean that $\mathcal{C}$ is a $\mathbb{k}$ -linear, abelian, rigid symmetric monoidal category with a tensor product $\otimes$ that is a bifunctor, a unit object $\mathbb{1}$ satisfying $\mathbb{1} \otimes \bullet = \text{Id}_{\mathcal{C}} = \bullet \otimes \mathbb{1}$, and an isomorphism $\text{End}_{\mathcal{C}}(\mathbb{1}) \cong \mathbb{k}$. For any object $X \in \text{Ob}(\mathcal{C})$, we write $\text{ev}_X: X^* \otimes X \rightarrow \mathbb{1}$ and $\text{coev}_X: \mathbb{1} \rightarrow X \otimes X^*$ for the evaluation and coevaluation maps, respectively (omitting subscripts when clear). For any objects X and $Y \in \text{Ob}(\mathcal{C})$, we denote the symmetric braiding isomorphism by $c_{X,Y}: X \otimes Y \xrightarrow{\sim} Y \otimes X$. We also use permutation notation for braiding isomorphisms: if σ is a permutation on n elements, then c_σ on an n -way tensor product denotes the braiding isomorphism which permutes the n components by σ (e.g., $c_{(123)}: X \otimes Y \otimes Z \rightarrow Z \otimes X \otimes Y$). The tensor product, as well as the evaluation, coevaluation, and braiding maps satisfy certain axioms and properties, all of which are described in detail in [9].

For any endomorphism $f \in \text{End}_{\mathcal{C}}(X)$, the *categorical trace* of f , denoted $\text{Tr}(f)$, is the endomorphism of the unit object given by the composition:

$$\text{Tr}(f) = \text{ev}_X \circ c_{X, X^*} \circ (f \otimes \text{Id}_{X^*}) \circ \text{coev}_X$$

Because $\text{End}_{\mathcal{C}}(\mathbb{1}) \cong \mathbb{k}$, the trace $\text{Tr}(f)$ is canonically identified with a scalar in the base field $\mathbb{k}$.

The *categorical dimension* of an object X , denoted $\dim(X)$, is defined simply as the categorical trace of its identity morphism:

$$\dim(X) = \text{Tr}(\text{Id}_X) = \text{ev}_X \circ c_{X, X^*} \circ \text{coev}_X$$

A *Lie algebra object* in $\mathcal{C}$ is a pair $(\mathfrak{g}, [\bullet, \bullet])$, where $\mathfrak{g} \in \text{Ob}(\mathcal{C})$ and $[\bullet, \bullet]: \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$ is a morphism in $\mathcal{C}$ (called the *commutator* or *bracket*) satisfying two conditions:

- (1) Anti-symmetry: $[\bullet, \bullet] \circ (\text{Id}_{\mathfrak{g} \otimes \mathfrak{g}} + c_{\mathfrak{g}, \mathfrak{g}}) = 0$.
- (2) Jacobi identity: $[\bullet, \bullet] \circ (\text{Id}_{\mathfrak{g}} \otimes [\bullet, \bullet]) \circ (\text{Id}_{\mathfrak{g} \otimes \mathfrak{g} \otimes \mathfrak{g}} + c_{(123)} + c_{(132)}^2) = 0$.

An *associative algebra object* in $\mathcal{C}$ is an object A equipped with an associative multiplication morphism $m: A \otimes A \rightarrow A$ and a unit morphism $u: \mathbb{1} \rightarrow A$. Any associative algebra object A naturally admits the structure of a Lie algebra object by defining the commutator bracket as $[\bullet, \bullet]_A = m \circ (\text{Id}_{A \otimes A} - c_{A, A})$.

The *universal enveloping algebra* of a Lie algebra object $\mathfrak{g}$, denoted $\mathcal{U}(\mathfrak{g})$, is an associative algebra object in $\mathcal{C}$ equipped with a morphism $i: \mathfrak{g} \rightarrow \mathcal{U}(\mathfrak{g})$ satisfying the relation $i \circ [\bullet, \bullet] = [\bullet, \bullet]_{\mathcal{U}(\mathfrak{g})} \circ (i \otimes i)$. Furthermore, it satisfies the universal property that for any associative algebra object A and any bracket-preserving morphism $f: \mathfrak{g} \rightarrow A$, there exists a unique associative algebra morphism $\tilde{f}: \mathcal{U}(\mathfrak{g}) \rightarrow A$ such that $\tilde{f} \circ i = f$. Constructively, $\mathcal{U}(\mathfrak{g})$ can be defined as the quotient of the tensor algebra $T(\mathfrak{g})$ by the two-sided ideal generated by the image of the morphism $\text{Id}_{\mathfrak{g} \otimes \mathfrak{g}} - c_{\mathfrak{g}, \mathfrak{g}} - [\bullet, \bullet]: \mathfrak{g} \otimes \mathfrak{g} \rightarrow T(\mathfrak{g})$.

For any object $V \in \text{Ob}(\mathcal{C})$, the *free Lie algebra* generated by V , denoted $FL(V)$, is a Lie algebra object in $\mathcal{C}$ equipped with a morphism $j: V \rightarrow FL(V)$ that satisfies the standard universal property: for any Lie algebra object $\mathfrak{h}$ in $\mathcal{C}$ and any morphism $f: V \rightarrow \mathfrak{h}$, there exists a unique Lie algebra morphism $\tilde{f}: FL(V) \rightarrow \mathfrak{h}$ such that $\tilde{f} \circ j = f$. The free Lie algebra $FL(V)$ can be realized explicitly as the smallest Lie subobject of the tensor algebra $T(V)$ (viewed as a Lie algebra object via the commutator bracket) that contains V . It follows from the universal properties that the universal enveloping algebra of the free Lie algebra is the tensor algebra.

2.3. Deligne Categories. We review the definition of Deligne categories from [1]. For a field $\mathbb{k}$ of characteristic 0, the Deligne category $\underline{\text{Rep}}(GL_t)$ for a parameter $t \in \mathbb{k}$ interpolates the classical representation category of the general linear group $GL_n(\mathbb{C})$ to non-integer dimensions. Rather than constructing modules over a group, $\underline{\text{Rep}}(GL_t)$ is defined as the idempotent completion (Karoubian envelope) of the additive closure of the free rigid monoidal $\mathbb{k}$ -linear category generated by an object V of categorical dimension t and its dual.

In classical GL_n representation theory, the finite-dimensional irreducible representations appearing in mixed tensor powers $V^{\otimes r} \otimes (V^*)^{\otimes s}$ are indexed by integer partitions $\lambda = (\lambda_1, \dots, \lambda_n)$, with $\lambda_1 \geq \dots \geq \lambda_n, \lambda_i \in \mathbb{Z}$. The indecomposable objects in $\underline{\text{Rep}}(GL_t)$ are parameterized by all bipartitions and are denoted $L_{\lambda, \mu}$ in the context of highest-weight modules.

The categorical structure of $\underline{\text{Rep}}(GL_t)$ depends strictly on whether the parameter t is an integer.

- **Generic Case** ($t \notin \mathbb{Z}$): When t is not an integer, the category $\underline{\text{Rep}}(GL_t)$ is abelian and semisimple. The indecomposable objects $L_{\lambda, \mu}$ are precisely the irreducible objects of the category.
- **Integer Specialization** ($t \in \mathbb{Z}$): When $t = n \in \mathbb{Z}$, the category ceases to be semisimple and becomes a Krull-Schmidt category. It develops a non-trivial tensor ideal of negligible morphisms $\mathcal{N}$ (morphisms f such that $\text{Tr}(f \circ g) = 0$ for all g). The classical representation category $\text{Rep}(GL_n(\mathbb{C}))$ is recovered by taking the quotient $\underline{\text{Rep}}(GL_n)/\mathcal{N}$. Under this quotient functor, the indecomposable objects $L_{\lambda, \mu}$ map to the standard classical irreducible representations if $l(\lambda) + l(\mu) \leq n$, and map to zero otherwise.

Example 2.2. The Deligne category $\underline{\text{Rep}}(GL_{\mathbf{n}_t})$ contains a Lie algebra object interpolating the general linear Lie algebra $\mathfrak{gl}_n$. When n is an integer, $\mathfrak{gl}_n$ is simply the $n \times n$ matrix algebra, which can be identified with $V \otimes V^*$ for an n -dimensional vector space V . Standard matrix multiplication is given by the map

$$1 \otimes \text{ev}_V \otimes 1 : (V \otimes V^*) \otimes (V \otimes V^*) \rightarrow V \otimes V^*.$$

The commutator in $\mathfrak{gl}_n$ is given by $[A, B] = AB - BA$, so it can be written as the map

$$(1 \otimes (\text{ev}_V \otimes 1) \otimes 1) \circ (\text{Id} - c_{(13)(24)}) : (V \otimes V^*) \otimes (V \otimes V^*) \rightarrow V \otimes V^*.$$

We can consider the same map on the object $V \otimes V^*$ in the Deligne category $\underline{\text{Rep}}(GL_{\mathbf{n}_t})$, which makes $V \otimes V^*$ a Lie algebra object which we call $\mathfrak{gl}_t$.

This object acts on the tautological module V the same way that it does in the integer case:

$$1 \otimes \text{ev}_V : (V \otimes V^*) \otimes V \rightarrow V,$$

and likewise on its dual V^* :

$$(-\text{ev}_V \otimes 1) \circ (c_{V \otimes V^*, V^*}) : (V \otimes V^*) \otimes V^* \rightarrow V^*.$$

Since a Lie algebra L naturally acts on the tensor products of modules via the usual coproduct $\Delta : L \rightarrow L \otimes L$, $\mathfrak{gl}_t$ acts on $L_{\lambda, \mu}$.

In the framework of this paper, we operate within the product of Deligne categories $\boxtimes_i \underline{\text{Rep}}(GL_{n_i})$ for parameters n_i . Each index i corresponds to an independent tautological object V_i with categorical dimension n_i .

3. AN ALTERNATIVE CONSTRUCTION OF KAC-MOODY LIE ALGEBRAS IN DELIGNE CATEGORIES

3.1. Associated Diagrams and Kac-Moody Lie Algebras. Let D be a weakly connected directed graph (allowing multiple edges and self-loops). Let $\mathbf{V}(D)$ and $\mathbf{E}(D)$ denote the vertices and edges of D , respectively, and let $\mathbf{L}(D)$ denote the set of edges which are self-loops. For a vertex $v \in \mathbf{V}(D)$, let S_v , T_v , and L_v denote the sets of outgoing, incoming, and self-loop edges at v , respectively, not including any self-loops in either S_v or T_v . Also let $S'_v = S_v \cup L_v$ and $T'_v = T_v \cup L_v$ denote the sets of *all* edges with v as the source or target, respectively. For an edge $e \in \mathbf{E}(D)$, let $s(e)$ and $t(e)$ denote its source and target, respectively.

Let $\mathbf{N} : \mathbf{E}(D) \rightarrow \mathbb{Z}_{\geq 2}$ be any function, and for $e \in \mathbf{E}(D)$, let $n_e = \mathbf{N}(e)$. From D and $\mathbf{N}$, we construct a new graph by inserting $n_e + 1$ consecutive vertices along e . We can treat this new graph as an undirected simply-laced Kac-Moody diagram, which we call the *associated diagram*. We let $v_{e,s}$ denote the first of these vertices (closest to the source $s(e)$) and $v_{e,t}$ denote the last; the remaining $n_e - 1$, which we call *intermediate vertices*, form a diagram of type A_{n_e-1} .

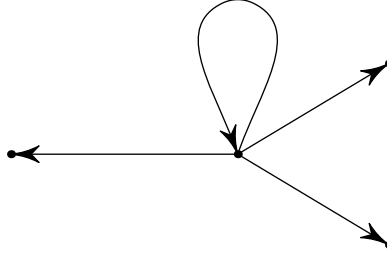


FIGURE 1. Oriented graph

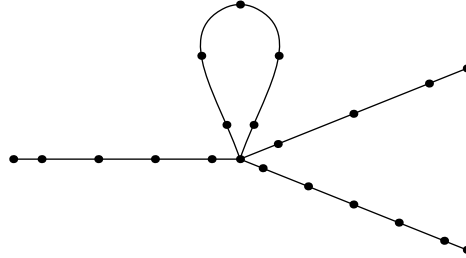


FIGURE 2. Graph with inserted vertices

Consider the Kac-Moody Lie algebra associated with the associated diagram. For a vertex $v \in \mathbf{V}(D)$, let e_v and f_v be the corresponding Chevalley generators. Likewise, for an edge $i \in \mathbf{E}(D)$, define $e_{i,s}$ and $f_{i,s}$ corresponding to $v_{i,s}$, and $e_{i,t}$ and $f_{i,t}$ corresponding to $v_{i,t}$.

For each edge $i \in \mathbf{E}(D)$, we add to this algebra a Cartan element Id_i which acts as follows:

$$[\text{Id}_i, e_{i,s}] = -e_{i,s}, \quad [\text{Id}_i, f_{i,s}] = f_{i,s}, \quad [\text{Id}_i, e_{i,t}] = e_{i,t}, \quad [\text{Id}_i, f_{i,t}] = -f_{i,t},$$

and Id_i acts as 0 on all other Chevalley generators. We denote the resulting algebra as $\mathfrak{g}(D, \mathbf{N})$.

In particular, Id_i along with the Chevalley generators for the $n_i - 1$ intermediate vertices along edge i generates a copy of $\mathfrak{gl}_{n_i}$ (with $\text{Id}_i \in \mathfrak{gl}_{n_i}$ representing the identity matrix, which is central in $\mathfrak{gl}_{n_i}$). For $1 \leq j, k \leq n_i$, we let $E_{j,k}^i$ denote the elementary matrix in $\mathfrak{gl}_{n_i}$ with (j, k) -th entry 1 and all other entries 0.

Also note that Id_i along with the generators for all $n_i + 1$ vertices along i generates a copy of $\mathfrak{gl}_{n_i+2}$, with the above copy of $\mathfrak{gl}_{n_i}$ in the middle n_i rows and columns. This is because the action of

Id_i on $e_{i,s}$, $f_{i,s}$, $e_{i,t}$, and $f_{i,t}$ matches how the identity matrix in $\mathfrak{gl}_{n_i}$ would act on the corresponding Chevalley generators in $\mathfrak{gl}_{n_i+2}$.

Let $\mathbf{V}'(D)$ denote the set of vertices including $\mathbf{V}(D)$ as well as $v_{i,s}$ and $v_{i,t}$ for all $i \in \mathbf{E}(D)$, and let $\Gamma(D)$ denote the free abelian group generated by $\mathbf{V}'(D)$. To the Chevalley generators e_v and f_v , we assign degrees v and $-v$, respectively, and likewise for $e_{i,s}$, $f_{i,s}$, $e_{i,t}$ and $f_{i,t}$. The remaining Chevalley generators are given degree 0. This makes $\mathfrak{g}(D, \mathbf{N})$ an algebra multi-graded by $\Gamma(D)$. For $\gamma \in \Gamma(D)$, let $\mathfrak{g}(D, \mathbf{N})_\gamma$ denote the degree- γ component of $\mathfrak{g}(D, \mathbf{N})$. Note that $\mathfrak{g}(D, \mathbf{N})_0$ contains $\mathfrak{gl}_{n_i}$ for all $i \in \mathbf{E}(D)$, along with $[e_{i,s}, f_{i,s}]$ and $[e_{i,t}, f_{i,t}]$ for $i \in \mathbf{E}(D)$ and $[e_v, f_v]$ for $v \in \mathbf{V}(D)$.

For an edge $i \in \mathbf{E}(D)$, let V_i denote the tautological $\mathfrak{gl}_{n_i}$ -module (with $\mathfrak{gl}_{n_j}$ acting as 0 for any edge $j \neq i$), and let V_i^* be its dual.

Proposition 3.1. *For any edge $i \in \mathbf{E}(D)$, the degree- $v_{i,s}$ component of $\mathfrak{g}(D, \mathbf{N})$ is*

$$\mathfrak{g}(D, \mathbf{N})_{v_{i,s}} \cong V_i^*$$

and the degree- $v_{i,t}$ component is

$$\mathfrak{g}(D, \mathbf{N})_{v_{i,t}} \cong V_i.$$

For any vertex $v \in \mathbf{V}(D)$, the degree- v component is

$$\mathfrak{g}(D, \mathbf{N})_v \cong \mathbb{1}.$$

We will refer to these components as $V_{i,s}^*$, $V_{i,t}$ and $\mathbb{1}_{v,+}$, and the corresponding negative-degree components as $V_{i,s}$, $V_{i,t}^*$, and $\mathbb{1}_{v,-}$.

Proof. The degree- $v_{i,s}$ component is generated by $e_{i,s}$. For any edge $j \neq i$, we have $\mathfrak{gl}_{n_j}$ acts as 0 on $e_{i,s}$ (as $v_{i,s}$ is not adjacent to any vertex along j), so it suffices to determine how $\mathfrak{gl}_{n_i}$ acts on $e_{i,s}$. This action generates the subspace of $\mathfrak{gl}_{n_i+2}$ with nonzero entries only in the middle n_i entries of the top row, which as a $\mathfrak{gl}_{n_i+2}$ -module is isomorphic to V_i^* .

The same argument holds for the degree- $v_{i,t}$ component, except $e_{i,t}$ generates the subspace of $\mathfrak{gl}_{n_i+2}$ with nonzero entries only in the middle n_i entries of the right column, which is isomorphic to V_i .

For any edge i , $\mathfrak{gl}_{n_i}$ acts trivially on e_v , so the degree- v component is simply $\mathbb{1}$. $\square$

For each edge $i \in \mathbf{E}(D)$ we can describe the Lie subalgebra generated by $\mathfrak{gl}_{n_i}$, $\mathbb{1}_{i,s}$, $\mathbb{1}_{i,t}$, $V_{i,s}^*$, $V_{i,t}$ and their duals in terms of the following matrix algebra (see the paragraph before [Corollary 3.6](#) for the definition of $\mathbb{1}_{i,\pm}$ and [Section 3.2](#) for the definition of $\mathbb{1}_{i,s}$ and $\mathbb{1}_{i,t}$):

$$\left(\begin{array}{c|c|c} \mathbb{1}_{i,s} & V_{i,s}^* & \mathbb{1}_{i,+} \\ \hline V_{i,s} & \mathfrak{gl}_{n_i} & V_{i,t} \\ \hline \mathbb{1}_{i,-} & V_{i,t}^* & \mathbb{1}_{i,t} \end{array} \right)$$

Thus, the positive part of $\mathfrak{g}(D, \mathbf{N})$ (the part generated by the positive-degree Chevalley generators) is the quotient of the free Lie algebra

$$(2) \quad FL \left(\bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}_{v,+} \oplus \bigoplus_{i \in \mathbf{E}(D)} V_{i,s}^* \oplus \bigoplus_{i \in \mathbf{E}(D)} V_{i,t} \right)$$

by the ideal I generated by the Serre relations. This also gives $\mathfrak{g}(D, \mathbf{N})$ a $\mathbb{Z}$ -grading where every vertex is given degree 1. We have already accounted for the Serre relations pertaining to the adjoint action of any intermediate vertex acting on e_v , $e_{i,s}$, and $e_{i,t}$ in [Proposition 3.1](#), as these tell us what modules the actions of $\mathfrak{gl}_{n_i}$ on these generators generate. Thus, I is generated by the following Serre relations:

- (1) $\text{ad}_{e_{i,s}}^2(E_{1,2}^i) = 0$ for $i \in \mathbf{E}(D)$, where $E_{1,2}^i$ is the positive Chevalley generator corresponding to the first root of $\mathfrak{gl}_{n_i}$.
- (2) $\text{ad}_{e_{i,t}}^2(E_{n_i-1,n_i}^i) = 0$ for $i \in \mathbf{E}(D)$, where E_{n_i-1,n_i}^i is the positive Chevalley generator corresponding to the last root of $\mathfrak{gl}_{n_i}$.
- (3) $[e_v, e_w] = 0$ for distinct $v, w \in \mathbf{V}(D)$.
- (4) $[e_{i,d_1}, e_{j,d_2}] = 0$ for distinct $i, j \in \mathbf{E}(D)$ and $d_1, d_2 \in \{s, t\}$.
- (5) $[e_{i,s}, e_{i,t}] = 0$ for $i \in \mathbf{E}(D)$.
- (6) $[e_v, e_{i,s}] = 0$ for $v \in \mathbf{V}(D)$ and $i \in \mathbf{E}(D)$ such that $v \neq s(i)$.
- (7) $[e_v, e_{i,t}] = 0$ for $v \in \mathbf{V}(D)$ and $i \in \mathbf{E}(D)$ such that $v \neq t(i)$.
- (8) $\text{ad}_{e_{s(i)}}^2(e_{i,s}) = \text{ad}_{e_{i,s}}^2(e_{s(i)}) = 0$ for $i \in \mathbf{E}(D)$.
- (9) $\text{ad}_{e_{t(i)}}^2(e_{i,t}) = \text{ad}_{e_{i,t}}^2(e_{t(i)}) = 0$ for $i \in \mathbf{E}(D)$.

Unlike in [13], this setup has degree-3 Serre relations: the last two relations above lie in the degree-3 component of Equation (2). Thus, it will be useful to analyze the degree-3 component of a free Lie algebra.

Lemma 3.2. *Let V be a vector space over a field k of characteristic 0. There are isomorphisms of $GL(V)$ -modules*

$$FL_2(V) \cong \Lambda^2 V$$

and

$$FL_3(V) \cong \mathbb{S}^{(2,1)} V,$$

where $FL_n(V)$ denotes the degree- n component of the free Lie algebra with respect to the standard grading and $\mathbb{S}^{(2,1)}$ denotes the Schur functor corresponding to the partition $(2, 1)$.

Proof. Consider the bracket map given by

$$\phi : V \otimes V \rightarrow FL_2(V), \quad v_1 \otimes v_2 \rightarrow [v_1, v_2].$$

The anti-symmetry of the Lie bracket implies that ϕ vanishes on symmetric tensors, hence factors uniquely through a surjective homomorphism of $GL(V)$ -modules $\bar{\phi} : \Lambda^2 V \rightarrow FL_2(V)$. Conversely, the inclusion map $i : FL_2(V) \hookrightarrow V^{\otimes 2}$ has image generated by all antisymmetric tensors. Defining $\psi_2 : FL_2(V) \rightarrow V \otimes V$ as

$$\psi_2([v_1, v_2]) = \frac{1}{2}(v_1 \otimes v_2 - v_2 \otimes v_1) = v_1 \wedge v_2,$$

we see that ψ_2 and $\bar{\phi}$ are inverses, so $FL_2(V) \cong \Lambda^2 V$.

The component $FL_3(V)$ is spanned by formal nested brackets of three elements: $[[v_1, v_2], v_3]$ and $[v_1, [v_2, v_3]]$. By the Jacobi identity,

$$[v_1, [v_2, v_3]] = [[v_1, v_2], v_3] - [[v_1, v_3], v_2] = [[v_1, v_2], v_3] + [[v_3, v_1], v_2],$$

so $FL_3(V)$ is generated by nested brackets of the form $[[v_1, v_2], v_3]$.

Since $[v_1, v_2] \in FL_2(V) \cong \Lambda^2 V$, we can define a surjective $GL(V)$ -equivariant linear map:

$$\pi : \Lambda^2 V \otimes V \longrightarrow FL_3(V), \quad (v_1 \wedge v_2) \otimes v_3 \longmapsto [[v_1, v_2], v_3].$$

To identify $FL_3(V)$, we decompose the domain $\Lambda^2 V \otimes V = \mathbb{S}^{(1,1)} V \otimes \mathbb{S}^{(1)} V$ using the Littlewood-Richardson rule:

$$\Lambda^2 V \otimes V \cong \mathbb{S}^{(2,1)} V \oplus \mathbb{S}^{(1,1,1)} V = \mathbb{S}^{(2,1)} V \oplus \Lambda^3 V.$$

Here, $\Lambda^3 V = \mathbb{S}^{(1,1,1)} V$ is embedded in $\Lambda^2 V \otimes V$ via the map

$$\iota : v_1 \wedge v_2 \wedge v_3 \longmapsto (v_1 \wedge v_2) \otimes v_3 + (v_2 \wedge v_3) \otimes v_1 + (v_3 \wedge v_1) \otimes v_2.$$

This map is well-defined, as

$$\begin{aligned}\iota(v_1 \wedge v_3 \wedge v_2) &= (v_1 \wedge v_3) \otimes v_2 + (v_3 \wedge v_2) \otimes v_1 + (v_2 \wedge v_1) \otimes v_3 \\ &= -(v_3 \wedge v_1) \otimes v_2 - (v_2 \wedge v_3) \otimes v_1 - (v_1 \wedge v_2) \otimes v_3 \\ &= -\iota(v_1 \wedge v_2 \wedge v_3),\end{aligned}$$

and likewise for other swaps.

Evaluating π on an element in the image of $\Lambda^3 V$,

$$\pi\left((v_1 \wedge v_2) \otimes v_3 + (v_2 \wedge v_3) \otimes v_1 + (v_3 \wedge v_1) \otimes v_2\right) = [[v_1, v_2], v_3] + [[v_2, v_3], v_1] + [[v_3, v_1], v_2] = 0,$$

which is identically zero by the Jacobi identity.

Thus, $\Lambda^3 V \subseteq \ker(\pi)$.

Since Schur functors $\mathbb{S}^\lambda V$ are irreducible polynomial $GL(V)$ -modules for distinct partitions λ , Schur's Lemma implies that the kernel $\ker(\pi)$ must be a sub-representation of $\mathbb{S}^{(2,1)} V \oplus \Lambda^3 V$. Because π is nonzero, $\ker(\pi)$ cannot contain $\mathbb{S}^{(2,1)} V$.

Therefore, $\ker(\pi) = \Lambda^3 V = \mathbb{S}^{(1,1,1)} V$, and by the First Isomorphism Theorem,

$$FL_3(V) \cong (\Lambda^2 V \otimes V) / \Lambda^3 V \cong \mathbb{S}^{(2,1)} V. \quad \square$$

Lemma 3.3. *Let $V = \bigoplus_{i=1}^n V_i$ be a direct sum of vector spaces over a field of characteristic 0. Then,*

$$\Lambda^2 V = \bigoplus_{i=1}^n \Lambda^2 V_i \oplus \bigoplus_{1 \leq i < j \leq n} V_i \otimes V_j$$

and

$$\mathbb{S}^{(2,1)} V = \bigoplus_{i=1}^n \mathbb{S}^{(2,1)} V_i \oplus \bigoplus_{i=1}^n \bigoplus_{\substack{j=1 \\ j \neq i}}^n S^2 V_i \otimes V_j \oplus \bigoplus_{i=1}^n \bigoplus_{\substack{j=1 \\ j \neq i}}^n \Lambda^2 V_i \otimes V_j \oplus \left(\bigoplus_{1 \leq i < j < k \leq n} V_i \otimes V_j \otimes V_k \right)^{\oplus 2}.$$

Proof. Both identities follow from repeated application of the Littlewood-Richardson rules for Schur functors of a direct sum. $\square$

Now we can compute the ideal I . Let

$$V = \bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}_{v,+} \oplus \bigoplus_{i \in \mathbf{E}(D)} V_{i,s}^* \oplus \bigoplus_{i \in \mathbf{E}(D)} V_{i,t}$$

be the degree-1 (generating) component of Equation (2). The degree-2 and degree-3 components are $\Lambda^2 V$ and $\mathbb{S}^{(2,1)} V$, respectively, by Lemma 3.2, which can be expanded via Lemma 3.3.

For $i \in \mathbf{E}(D)$ and $d \in \{s, t\}$, let $V_{i,d}^{(*)}$ denote $V_{i,s}^*$ if $d = s$ and $V_{i,t}$ if $d = t$ (i.e., the degree- $e_{i,d}$ part of $\mathfrak{g}(D, \mathbf{N})$). Also let $d(i)$ denote $s(i)$ if $d = s$ and $t(i)$ if $d = t$ (i.e., the vertex in $\mathbf{V}(D)$ adjacent to $v_{i,d}$).

Proposition 3.4. *The degree-2 component of the ideal I is the direct sum of the following subspaces:*

- (1) $\Lambda^2 V_{i,d}^{(*)}$ for $i \in \mathbf{E}(D)$ and $d \in \{s, t\}$.
- (2) $\mathbb{1}_{v,+} \otimes \mathbb{1}_{w,+}$ for distinct $v, w \in \mathbf{V}(D)$.
- (3) $V_{i,d_1}^{(*)} \otimes V_{j,d_2}^{(*)}$ for distinct $i, j \in \mathbf{E}(D)$ and $d_1, d_2 \in \{s, t\}$.
- (4) $\mathbb{1}_{v,+} \otimes V_{i,d}^{(*)}$ for $v \in \mathbf{V}(D)$, $i \in \mathbf{E}(D)$, and $d \in \{s, t\}$ such that $v \neq d(i)$.
- (5) $L_{i,+,[1],(1)}$ for $i \in \mathbf{E}(D)$, denoting the kernel of the evaluation map $V_{i,s}^* \otimes V_{i,t} \rightarrow \mathbb{1}$.

Proof. It suffices to determine which subspace of the expansion of $\Lambda^2 V$ each Serre relation lies in. Recall that $V_{i,s}^*$ and $V_{i,t}$ both lie in $\mathfrak{gl}_{n_i+2}$, with $V_{i,s}^*$ having nonzero entries in the middle n_i entries of the top row, and $V_{i,t}$ having nonzero entries in the middle n_i entries of the right column.

- (1) For $i \in \mathbf{E}(D)$, the relation $\text{ad}_{e_{i,s}}^2(E_{1,2}^i)$ lies in $\Lambda^2 V_{i,s}^*$, as it has degree $2v_{i,s}$. Thus, it generates $\Lambda^2 V_{i,s}^*$. (It is also apparent that $[V_{i,s}^*, V_{i,s}^*] = 0$ from the above identification of $V_{i,s}^*$.)
Similarly, $\text{ad}_{e_{i,t}}^2(E_{n_i-1,n_i}^i)$ lies in $\Lambda^2 V_{i,t}$.
- (2) For distinct $v, w \in \mathbf{V}(D)$, the relation $[e_v, e_w] = 0$ lies in $\mathbb{1}_{v,+} \otimes \mathbb{1}_{w,+}$ and thus generates it.
- (3) For distinct $i, j \in \mathbf{E}(D)$ and $d_1, d_2 \in \{s, t\}$, the relation $[e_{i,d_1}, e_{j,d_2}] = 0$ lies in $V_{i,d_1}^{(*)} \otimes V_{j,d_2}^{(*)}$ and thus generates it.
- (4) For $v \in \mathbf{V}(D)$, $i \in \mathbf{E}(D)$, and $d \in \{s, t\}$ such that $v \neq d(i)$, the vertices v and $v_{i,d}$ are not adjacent. The relation $[e_v, e_{i,d}] = 0$ lies in $\mathbb{1}_{v,+} \otimes V_{i,d}^{(*)}$ and thus generates it.
- (5) The relation $[e_{i,s}, e_{i,t}] = 0$ lies in $V_{i,s}^* \otimes V_{i,t}$. However, $V_i^* \otimes V_i$ is not irreducible, so the relation may not generate all of it.

Consider an element $x \in V_{i,s}^*$ and an element $y \in V_{i,t}$. By the identifications above as subspaces of $\mathfrak{gl}_{n_i+2}$, we can express these elements as matrices

$$x = \begin{pmatrix} 0 & x_1 & \cdots & x_{n_i} & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix} \quad \text{and} \quad y = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & y_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & y_{n_i} \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Then

$$[x, y] = \begin{pmatrix} 0 & 0 & \cdots & 0 & x_1 y_1 + \cdots + x_{n_i} y_{n_i} \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix},$$

so the commutator map between $V_{i,s}^*$ and $V_{i,t}$ is given by the evaluation map into the top-right entry of $\mathfrak{gl}_{n_i+2}$. Thus, the relation $[e_{i,s}, e_{i,t}] = 0$ generates the kernel of the evaluation map.

This accounts for all degree-2 Serre relations, so the proposition is proved. $\square$

Proposition 3.5. *The degree-3 component of the ideal I is generated by the degree-2 component as in Proposition 3.4, plus the direct sum of the following subspaces:*

- (6) $S^2 \mathbb{1}_{d(i),+} \otimes V_{i,d}^{(*)}$ for $i \in \mathbf{E}(D)$.
- (7) $S^2 V_{i,d}^{(*)} \otimes \mathbb{1}_{d(i),+}$ for $i \in \mathbf{E}(D)$.

Proof. First consider the degree-3 component of a free Lie algebra $FL(V \oplus W)$ for vector spaces V and W . By Lemma 3.2, it is

$$FL_3(V \oplus W) = \mathbb{S}^{(2,1)}(V \oplus W) = \mathbb{S}^{(2,1)}V \oplus \mathbb{S}^{(2,1)}W \oplus S^2 V \otimes W \oplus S^2 W \otimes V \oplus \Lambda^2 V \otimes W \oplus \Lambda^2 W \otimes V.$$

Consider the map $\phi : V \otimes V \otimes W \rightarrow FL_3(V \oplus W)$ given by

$$v_1 \otimes v_2 \otimes w \mapsto [v_1, [v_2, w]].$$

It is injective (this can be verified over bases of V and W) and its image lies in the part of $FL_3(V \oplus W)$ with degree 2 in V and 1 in W , i.e.,

$$S^2 V \otimes W \oplus \Lambda^2 V \otimes W \cong V \otimes V \otimes W.$$

Thus, ϕ maps $S^2 V \otimes W \subseteq V \otimes V \otimes W$ into $S^2 V \otimes W \subseteq FL_3(V \oplus W)$. As $S^2 V \otimes W \subseteq V \otimes V \otimes W$ is generated by elements $v \otimes v \otimes w$ for $v \in V$ and $w \in W$, we conclude $S^2 V \otimes W \subseteq FL_3(V \oplus W)$ is generated by elements $[v, [v, w]] = \text{ad}_v^2 w$. Similarly, $S^2 W \otimes V$ is generated by elements $\text{ad}_w^2 v$.

Thus, for $i \in \mathbf{E}(D)$ and $d \in \{s, t\}$, the relation $\text{ad}_{e_{d(i)}}^2(e_{i,d}) = 0$ lies in $S^2\mathbb{1}_{d(i),+} \otimes V_{i,d}^{(*)}$ and thus generates it. Likewise, the relation $\text{ad}_{e_{i,d}}^2(e_{d(i)})$ lies in $S^2V_{i,d}^{(*)} \otimes \mathbb{1}_{d(i),+}$ and thus generates it. This accounts for all degree-3 Serre relations, so the proposition is proved. $\square$

Using [Proposition 3.4](#) and [Proposition 3.5](#), we can compute the degree-2 and degree-3 components of $\mathfrak{g}(D, \mathbf{N})$. For an edge $i \in \mathbf{E}(D)$, let $\mathbb{1}_{i,+}$ denote the image of the evaluation map $\text{ev} : V_{i,s}^* \otimes V_{i,t} \rightarrow \mathbb{1}$, i.e., the quotient of $V_{i,s}^* \otimes V_{i,t}$ by $L_{i,+,[(1), (1)]}$. Also let W_a denote a term of the form either $\mathbb{1}_{v,+}$ for $v \in \mathbf{V}(D)$ or $V_{i,d}^{(*)}$ for $i \in \mathbf{E}(D)$ and $d \in \{s, t\}$, with A being the index set so that $V = \bigoplus_{a \in A} W_a$.

Corollary 3.6. *The degree-2 component of $\mathfrak{g}(D, \mathbf{N})$ is*

$$\mathfrak{g}(D, \mathbf{N})_2 = \left(\bigoplus_{i \in \mathbf{E}(D)} \mathbb{1}_{s(i),+} \otimes V_{i,s}^* \right) \oplus \left(\bigoplus_{i \in \mathbf{E}(D)} \mathbb{1}_{t(i),+} \otimes V_{i,t} \right) \oplus \left(\bigoplus_{i \in \mathbf{E}(D)} \mathbb{1}_{i,+} \right).$$

Proof. By [Lemma 3.2](#) and [Lemma 3.3](#), the degree-2 component of [Equation \(2\)](#) is

$$(3) \quad \bigoplus_{a \in A} \Lambda^2 W_a \oplus \bigoplus_{\{a,b\} \in \binom{A}{2}} W_a \otimes W_b,$$

where the second sum is over unordered pairs $\{a, b\}$. Thus, $\mathfrak{g}(D, \mathbf{N})_2$ is the quotient of [Equation \(3\)](#) by the subspaces in [Proposition 3.4](#). To determine which terms remain after this quotient, we casework first on the type of term (as in the expansion in [Equation \(3\)](#)), then on whether the constituent W_a 's are of the form $\mathbb{1}_{v,+}$ or $V_{i,d}^{(*)}$.

- We first consider terms of the form $\Lambda^2 W_a$.
 - If $W_a = \mathbb{1}_{v,+}$ for $v \in \mathbf{V}(D)$, then $\Lambda^2 W_a = 0$ because $\Lambda^2 \mathbb{1} = 0$, so it contributes nothing to $\mathfrak{g}(D, \mathbf{N})_2$.
 - If $W_a = V_{i,d}^{(*)}$ for $i \in \mathbf{E}(D)$ and $d \in \{s, t\}$, then $\Lambda^2 W_a$ is in subspace (1) of [Proposition 3.4](#), so it contributes nothing to $\mathfrak{g}(D, \mathbf{N})_2$.
- We next consider terms of the form $W_a \otimes W_b$.
 - If $W_a = \mathbb{1}_{v,+}$ and $W_b = \mathbb{1}_{w,+}$ for $v, w \in \mathbf{V}(D)$, then $W_a \otimes W_b$ is in subspace (2) of [Proposition 3.4](#), so it contributes nothing to $\mathfrak{g}(D, \mathbf{N})_2$.
 - If $W_a = \mathbb{1}_{v,+}$ and $W_b = V_{i,d}^{(*)}$ for $v \in \mathbf{V}(D)$, $i \in \mathbf{E}(D)$, and $d \in \{s, t\}$, then $W_a \otimes W_b$ is in subspace (4) of [Proposition 3.4](#) if and only if $v \neq d(i)$. Thus, the only contributions to $\mathfrak{g}(D, \mathbf{N})_2$ from these terms are $\mathbb{1}_{v,+} \otimes V_{i,d}^{(*)}$ for $i \in \mathbf{E}(D)$ and $d \in \{s, t\}$.
 - If $W_a = V_{i,d_1}^{(*)}$ and $W_b = V_{j,d_2}^{(*)}$ for $i, j \in \mathbf{E}(D)$ and $d_1, d_2 \in \{s, t\}$, then $W_a \otimes W_b$ is in subspace (3) of [Proposition 3.4](#) if and only if $i \neq j$, in which case it contributes nothing to $\mathfrak{g}(D, \mathbf{N})_2$. Otherwise, $i = j$, so the contribution to $\mathfrak{g}(D, \mathbf{N})$ is the quotient $V_{i,s}^* \otimes V_{i,t} / L_{i,+,[(1), (1)]} = \mathbb{1}_{i,+}$ by subspace (5) of [Proposition 3.4](#).

This exhausts all cases, so

$$\mathfrak{g}(D, \mathbf{N})_2 = \bigoplus_{i \in \mathbf{E}(D)} \bigoplus_{d \in \{s, t\}} \mathbb{1}_{d(i),+} \otimes V_{i,d}^{(*)} \oplus \bigoplus_{i \in \mathbf{E}(D)} \mathbb{1}_{i,+},$$

exactly as claimed. $\square$

Corollary 3.7. *The degree-3 component of $\mathfrak{g}(D, \mathbf{N})$ is the direct sum of the following subspaces:*

- (1) $(\mathbb{1}_{t(i),+} \otimes V_{i,s}^* \otimes V_{i,t}) \oplus (\mathbb{1}_{t(i),+} \otimes \mathbb{1}_{i,+})$ for all $i \in \mathbf{L}(D)$ (note $s(i) = t(i)$).
- (2) $\mathbb{1}_{s(i),+} \otimes \mathbb{1}_{i,+}$ for all $i \in \mathbf{E}(D) \setminus \mathbf{L}(D)$.
- (3) $\mathbb{1}_{t(i),+} \otimes \mathbb{1}_{i,+}$ for all $i \in \mathbf{E}(D) \setminus \mathbf{L}(D)$.
- (4) $\mathbb{1}_{v,+} \otimes V_{i,s}^* \otimes V_{j,s}$ for all $v \in \mathbf{V}(D)$ and unordered pairs of elements $i, j \in S'_v$.

- (5) $\mathbb{1}_{v,+} \otimes V_{i,t} \otimes V_{j,t}$ for all $v \in \mathbf{V}(D)$ and unordered pairs of elements $i, j \in T'_v$.
 (6) $\mathbb{1}_{v,+} \otimes V_{i,s}^* \otimes V_{j,t}$ for all $v \in \mathbf{V}(D)$, $i \in S'_v$, and $j \in T'_v$ with $i \neq j$.

Proof. By Lemma 3.2 and Lemma 3.3, the degree-3 component of Equation (2) is

$$(4) \quad \bigoplus_{a \in A} \mathbb{S}^{(2,1)} W_a \oplus \bigoplus_{a \in A} \bigoplus_{\substack{b \in A \\ b \neq a}} S^2 W_a \otimes W_b \oplus \bigoplus_{a \in A} \bigoplus_{\substack{b \in A \\ b \neq a}} \Lambda^2 W_a \otimes W_b \oplus \left(\bigoplus_{\{a,b,c\} \in \binom{A}{3}} W_a \otimes W_b \otimes W_c \right)^{\oplus 2}.$$

Thus, $\mathfrak{g}(D, \mathbf{N})_3$ is the quotient of Equation (4) by the subspaces in Proposition 3.5. To determine which terms remain after this quotient, we casework on the type of term (as in the expansion in Equation (4)).

- We first consider terms of the form $\mathbb{S}^{(2,1)} W_a$. None of these can contribute anything to $\mathfrak{g}(D, \mathbf{N})_3$, as the degree-2 component does not contain $\Lambda^2 W_a$ (Corollary 3.6).
- We next consider terms of the form $S^2 W_a \otimes W_b$. If $[W_a, W_b] = 0$, then this relation generates $S^2 W_a \otimes W_b$, so it contributes nothing to $\mathfrak{g}(D, \mathbf{N})_3$. Thus, it suffices to consider cases where $[W_a, W_b] \neq 0$.

- If $W_a = \mathbb{1}_{d(i),+}$ and $W_b = V_{i,d}^{(*)}$ for $i \in \mathbf{E}(D)$ and $d \in \{s, t\}$, then $S^2 W_a \otimes W_b$ is in subspace (6) of Proposition 3.5, so it contributes nothing to $\mathfrak{g}(D, \mathbf{N})$.
- If $W_a = V_{i,d}^{(*)}$ and $W_b = \mathbb{1}_{d(i),+}$ for $i \in \mathbf{E}(D)$ and $d \in \{s, t\}$, then $S^2 W_a \otimes W_b$ is in subspace (7) of Proposition 3.5, so it contributes nothing to $\mathfrak{g}(D, \mathbf{N})$.
- If $W_a = V_{i,s}^*$ and $W_b = V_{i,t}$ for $i \in \mathbf{E}(D)$, recall that $S^2 W_a \otimes W_b$ is generated by commutators of the form $[x, [x, y]]$ for $x \in W_a$ and $y \in W_b$. In this case, $[V_{i,s}^*, V_{i,t}] = \mathbb{1}_{i,+}$ by (5) of Proposition 3.4, so such commutators are always in $V_{i,s}^* \otimes \mathbb{1}_{i,+}$. However, we can show that all such commutators also vanish. For any $x \in V_{i,s}^*$, let $z \in V_{i,t}$ such that $\text{ev}(x \otimes z) = 0$, and let $y \in V_{i,s}^*$ such that $\text{ev}(y \otimes z) = 1$. (Such z exists because we assumed $n_i \geq 2$.) Then, $[x, y] = 0$ because commutators within $V_{i,s}^*$ all vanish. We also have $[x, z] = 0$ and $[y, z] = \mathbb{1}_{i,+}$ (denoting $1 \in \mathbb{1}_{i,+}$), as the commutator between $V_{i,s}^*$ and $V_{i,t}$ is given by evaluation. By the Jacobi identity,

$$0 = [[x, y], z] = [x, [y, z]] - [y, [x, z]] = [x, \mathbb{1}_{i,+}],$$

so $[x, \mathbb{1}_{i,+}] = 0$ for all $x \in V_{i,s}^*$.

We conclude that in this case, there is no contribution to $\mathfrak{g}(D, \mathbf{N})_3$.

- If $W_a = V_{i,t}$ and $W_b = V_{i,s}^*$ for $i \in \mathbf{E}(D)$, a similar argument shows there is no contribution to $\mathfrak{g}(D, \mathbf{N})_3$.

This exhausts all cases where $[W_a, W_b] \neq 0$.

- We next consider terms of the form $\Lambda^2 W_a \otimes W_b$. None of these can contribute anything to $\mathfrak{g}(D, \mathbf{N})_3$, as the degree-2 component does not contain $\Lambda^2 W_a$ (Corollary 3.6).
- Finally, we consider terms of the form $W_a \otimes W_b \otimes W_c$. Note that Equation (4) has two copies of this term. So long as at least *two* of the three relations $[W_a, W_b] = 0$, $[W_b, W_c] = 0$, and $[W_c, W_a] = 0$ hold, this term contributes nothing to $\mathfrak{g}(D, \mathbf{N})_3$. For instance, if $[W_a, W_b] = 0$ and $[W_b, W_c] = 0$, then W_b commutes with both W_a and W_c , and hence $[W_a, W_c]$ as well, so every three-way commutator between W_a , W_b , and W_c vanishes.

We now casework on how many of W_a , W_b , and W_c are of the form $\mathbb{1}_{v,+}$ for $v \in \mathbf{V}(D)$.

- If $W_a = \mathbb{1}_{v,+}$, $W_b = \mathbb{1}_{w,+}$, and $W_c = \mathbb{1}_{x,+}$ for $v, w, x \in \mathbf{V}(D)$, then $[W_a, W_b] = [W_b, W_c] = [W_c, W_a] = 0$ by (2) of Proposition 3.4, so in this case there is no contribution to $\mathfrak{g}(D, \mathbf{N})_3$.
- If $W_a = \mathbb{1}_{v,+}$, $W_b = \mathbb{1}_{w,+}$, and $W_c = V_{i,d}^{(*)}$ for $v, w \in \mathbf{V}(D)$, $i \in \mathbf{E}(D)$, and $d \in \{s, t\}$, then $[W_a, W_b] = 0$ by (2) of Proposition 3.4. Furthermore, by (4) of Proposition 3.4,

- $[W_a, W_c] \neq 0$ if and only if $v = d(i)$, and $[W_b, W_c] \neq 0$ if and only if $w = d(i)$. As $v \neq w$, at most one of these can be true, so at least one of $[W_a, W_c] = 0$ and $[W_b, W_c] = 0$ holds. Thus, in this case, there is no contribution to $\mathfrak{g}(D, \mathbf{N})_3$.
- If $W_a = \mathbb{1}_{v,+}$, $W_b = V_{i,d_1}^{(*)}$, and $W_c = V_{j,d_2}^{(*)}$ for $v \in \mathbf{V}(D)$, $i, j \in \mathbf{E}(D)$, and $d_1, d_2 \in \{s, t\}$, then by (3) and (4) of [Proposition 3.4](#),
 - * $[W_a, W_b] \neq 0$ if and only if $v = d_1(i)$.
 - * $[W_a, W_c] \neq 0$ if and only if $v = d_2(j)$.
 - * $[W_b, W_c] \neq 0$ if and only if $i = j$.
 Thus, it suffices to consider only cases where at least two of these three conditions hold. We casework on whether $i = j$.
 - * First suppose $i = j$, so the other two conditions become $v = s(i)$ and $v = t(i)$. By (5) of [Proposition 3.4](#), $[W_b, W_c] = \mathbb{1}_{i,+}$.
 - If $v = s(i) = t(i)$, then i is a self-loop. In this case, there are no relations between W_a and W_b , nor between W_a and W_c . Thus, the contribution to $\mathfrak{g}(D, \mathbf{N})_3$ in this case is the quotient of $(\mathbb{1}_{v,+} \otimes V_{i,s}^* \otimes V_{i,t})^{\oplus 2}$ by commutators of the form $[x, [y, z]]$ for $x \in \mathbb{1}_{v,+}$, $y \in V_{i,s}^*$, and $z \in V_{i,t}$ such that $y \otimes z \in L_{i,+,[1),(1)]}$. Such commutators generate $\mathbb{1}_{v,+} \otimes L_{i,+,[1),(1)]}$, so the contribution is

$$(\mathbb{1}_{v,+} \otimes V_{i,s}^* \otimes V_{i,t})^{\oplus 2} / (\mathbb{1}_{v,+} \otimes L_{i,+,[1),(1)]}) = (\mathbb{1}_{v,+} \otimes V_{i,s}^* \otimes V_{i,t}) \oplus (\mathbb{1}_{v,+} \otimes \mathbb{1}_{i,+}).$$
 - If $v = s(i) \neq t(i)$, then i is not a self-loop. Then $[\mathbb{1}_{v,+}, V_{i,t}] = 0$ and $[V_{i,s}^*, V_{i,t}] = \mathbb{1}_{i,+}$. Thus, the contribution to $\mathfrak{g}(D, \mathbf{N})_3$ in this case is $\mathbb{1}_{v,+} \otimes \mathbb{1}_{i,+}$.
 - If $v = t(i) \neq s(i)$, a similar argument also shows the contribution to $\mathfrak{g}(D, \mathbf{N})_3$ in this case is $\mathbb{1}_{v,+} \otimes \mathbb{1}_{i,+}$.
 - * Now suppose $i \neq j$, so we must have $v = d_1(i) = d_2(j)$. In this case, there are no relations between W_a and W_b , nor between W_a and W_c . As $i \neq j$, $[W_b, W_c] = 0$ by (3) of [Proposition 3.4](#). Thus, the contribution in this case is $\mathbb{1}_{v,+} \otimes V_{i,d_1}^{(*)} \otimes V_{j,d_2}^{(*)}$. - If $W_a = V_{i,d_1}^{(*)}$, $W_b = V_{j,d_2}^{(*)}$, and $W_c = V_{k,d_3}^{(*)}$ for $i, j, k \in \mathbf{E}(D)$ and $d_1, d_2, d_3 \in \{s, t\}$, then by (3) of [Proposition 3.4](#), $[W_a, W_b] \neq 0$ if and only if W_a and W_b are $V_{i,s}^{(*)}$ and $V_{i,t}$ in some order for some edge $i \in \mathbf{E}(D)$. The same holds for $[W_b, W_c]$ and $[W_c, W_a]$. As W_a, W_b , and W_c are distinct, there can be at most one pair of duals among them. Thus, at least two of the pairwise commutators vanish, so in this case there is no contribution to $\mathfrak{g}(D, \mathbf{N})_3$.

This exhausts all cases, and the terms in $\mathfrak{g}(D, \mathbf{N})_3$ are exactly as claimed. $\square$

The negative part of $\mathfrak{g}(D, \mathbf{N})$ admits a similar description with the dual terms.

3.2. Coroots of the Associated Algebra. [Section 3.1](#) classifies the object we want to interpolate, but we also need to be able to specify the commutator map on this object. As in [\[13\]](#), it suffices to specify the pairwise commutators between the degree-(-1), degree-0, and degree-1 components of $\mathfrak{g}(D, \mathbf{N})$; the full commutator map can then be recovered via the Jacobi identity.

Let us first specify the degree-0 component. It contains $\mathfrak{gl}_{n_e}$ for every edge $e \in \mathbf{E}(D)$. In addition, for all $i \in \mathbf{E}(D)$, there exists a degree-0 element $1_{i,s}$ such that

$$[1_{i,s}, e_{i,s}] = e_{i,s}, \quad [1_{i,s}, e_{s(i)}] = -e_{s(i)},$$

and $1_{i,s}$ acts as 0 on all other positive Chevalley generators. Likewise, there exists a degree-0 element $1_{i,t}$ such that

$$[1_{i,t}, e_{i,t}] = e_{i,t}, \quad [1_{i,t}, e_{t(i)}] = -e_{t(i)},$$

and $1_{i,t}$ acts as 0 on all other positive Chevalley generators. The reason we define $1_{i,s}$ and $1_{i,t}$ as such is that they can be realized as elements of $\mathfrak{gl}_{n_e+2}$ this way: $1_{i,s}$ is the elementary matrix with top-left entry 1, and $1_{i,t}$ is *negative* the elementary matrix with bottom-right entry 1.

For all $v \in \mathbf{V}(D)$, there exists a degree-0 element 1_v such that

$$[1_v, e_v] = e_v$$

and 1_v acts as 0 on all other positive Chevalley generators. Then, the degree-0 part of $\mathfrak{g}(D, \mathbf{N})$ is spanned by $\mathfrak{gl}_{n_i}$, $1_{i,s}$, and $1_{i,t}$ for all $i \in \mathbf{E}(D)$, 1_v for all $v \in \mathbf{V}(D)$, and possibly some additional central elements (e.g., if $\mathfrak{g}(D, \mathbf{N})$ is affine).

To determine the commutator between degree 1 and degree -1 , recall that the commutator between $\mathfrak{g}(D, \mathbf{N})_v$ and $\mathfrak{g}(D, \mathbf{N})_{-w}$ for vertices $v, w \in \mathbf{V}'(D)$ is 0. Thus, it suffices to specify $[e_{i,s}, f_{i,s}]$ and $[e_{i,t}, f_{i,t}]$ for $i \in \mathbf{E}(D)$, as well as $[e_v, f_v]$ for $v \in \mathbf{V}(D)$. These are the corresponding simple coroots $\alpha_{i,s}^\vee$, $\alpha_{i,t}^\vee$, and $\alpha_v^\vee$.

As we work in the simply-laced case, the coroot corresponding to a vertex v can be classified up to a central element by how it acts on the positive Chevalley generators: 2 on e_v , -1 on e_w for w adjacent to v , and 0 otherwise. Thus, $\alpha_{i,s}^\vee$ acts as 2 on $e_{i,s}$, -1 on both $e_{s(i)}$ and $E_{1,2}^i$ (the positive Chevalley generator corresponding to the first intermediate vertex along i), and 0 on all other positive Chevalley generators. Thus,

$$\alpha_{i,s}^\vee = 1_{i,s} - E_{1,1}^i,$$

up to a central element. Similarly,

$$\alpha_{i,t}^\vee = 1_{i,t} + E_{n_i, n_i}^i.$$

Note that these are also the corresponding coroots in $\mathfrak{gl}_{n_i+2}$.

For $v \in \mathbf{V}(D)$, v is adjacent to $v_{i,s}$ for $i \in S'_v$ and $v_{i,t}$ for $i \in T'_v$, so $\alpha_v^\vee$ acts as -1 on the corresponding Chevalley generators. Thus, we should be able to write

$$\alpha_v^\vee = - \left(\sum_{i \in S'_v} 1_{i,s} \right) - \left(\sum_{i \in T'_v} 1_{i,t} \right) + c_v 1_v$$

up to a central element for some coefficient c_v . (In fact, if $\mathfrak{g}(D, \mathbf{N})$ is an affine Lie algebra, then the central element does appear in the coroot $\alpha_v^\vee$.) As $\alpha_v^\vee$ acts as 2 on e_v , and each $1_{i,s}$ and $1_{i,t}$ in the sum above acts as -1 , we conclude $c_v = 2 - N_v$ where $N_v = |S'_v| + |T'_v|$ is the sum of the in-degree and out-degree of v (which counts self-loops at v twice).

This specifies the coroots, and thus the commutators between degree 1 and degree -1 . The commutators between degree 0 and degree ± 1 are already specified by definition.

3.3. Interpolation of Kac-Moody Lie Algebras. With the descriptions in [Section 3.1](#) and [Section 3.2](#), we are ready to interpolate this construction of $\mathfrak{g}(D, \mathbf{N})$ from an integer-valued function $\mathbf{N}$ to a complex-valued function.

Definition 3.8. Let D be a weakly connected directed graph. Let $\delta : \mathbf{E}(D) \rightarrow \mathbb{C}$ be a complex-valued function. For $e \in \mathbf{E}(D)$, let $\delta_e = \delta(e)$, and suppose the values of δ_e for all $e \in \mathbf{E}(D)$ are transcendental and algebraically independent.

Consider the product of Deligne categories $\mathcal{C} := \boxtimes_{e \in \mathbf{E}(D)} \underline{\text{Rep}}(GL_{\mathbf{n}_{\delta_e}})$. For $e \in \mathbf{E}(D)$, let V_e denote the tautological module in $\underline{\text{Rep}}(GL_{\mathbf{n}_{\delta_e}})$, and let V_e^* denote its dual. ($V_{e,s}$ and $V_{e,t}$ denote objects isomorphic to V_e , and so on.) We define the [categorical Kac-Moody Lie algebra](#) $\text{KM}(D, \delta)$ as an ind-object of this category:

$$\text{KM}(D, \delta) = \mathbf{u}_- \oplus \mathbf{l} \oplus \mathbf{u}_+,$$

where

$$\mathfrak{l} = \left(\bigoplus_{i \in \mathbf{E}(D)} V_i \otimes V_i^* \right) \oplus \left(\bigoplus_{i \in \mathbf{E}(D)} \mathbb{1}_{i,s} \right) \oplus \left(\bigoplus_{i \in \mathbf{E}(D)} \mathbb{1}_{i,t} \right) \oplus \left(\bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}_v \right) \oplus \left(\bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}'_v \right)$$

is the Levi subalgebra, the positive part $\mathfrak{u}_+$ is given by the quotient of the free Lie algebra Equation (2) (well-defined by [8]) by the ideal generated by the relations in Proposition 3.4 and Proposition 3.5, and the negative part $\mathfrak{u}_-$ is the quotient of the dual free Lie algebra by the dual ideal. (The Levi subalgebra $\mathfrak{l}$ is given additional central parts $\mathbb{1}'_v$ for each vertex $v \in \mathbf{V}(D)$; this is to capture behavior such as in the case of affine Lie algebras, and is otherwise typically a trivial central extension.)

We now specify the commutator between degree 1 and degree -1 .

- For $i \in \mathbf{E}(D)$, the commutator between degree $v_{i,s}$ and degree $-v_{i,s}$ is given by the map

$$\text{ev}_{V_i} \oplus (-c_{V_i^*, V_i}) : V_{i,s}^* \otimes V_{i,s} \rightarrow \mathbb{1}_{i,s} \oplus (V_i \otimes V_i^*) \subseteq \mathfrak{l}.$$

- Likewise, the commutator between degree $v_{i,t}$ and degree $-v_{i,t}$ is given by the map

$$\text{Id} \oplus \text{ev}_{V_i^*} : V_{i,t} \otimes V_{i,t}^* \rightarrow (V_i \otimes V_i^*) \oplus \mathbb{1}_{i,t} \subseteq \mathfrak{l}.$$

(Recall that in the integer case, $\mathbb{1}_{i,t}$ is negative the lower-right elementary matrix of $\mathfrak{gl}_{n_i+2}$, which is why this commutator map has ev_{V_i} and not $-\text{ev}_{V_i}$.)

- For $v \in \mathbf{V}(D)$, the commutator between degree v and degree $-v$ is the map

$$\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow \mathbb{1}'_v \oplus \left(\bigoplus_{i \in S'_v} \mathbb{1}_{i,s} \right) \oplus \left(\bigoplus_{i \in T'_v} \mathbb{1}_{i,t} \right) \oplus \mathbb{1}_v \in \mathfrak{l}$$

which is the direct sum of the maps

- the multiplication map $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow \mathbb{1}'_v$,
- (-1) times the multiplication map $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow \mathbb{1}_{i,s}$ for all $i \in S'_v$,
- (-1) times the multiplication map $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow \mathbb{1}_{i,t}$ for all $i \in T'_v$, and
- $(2 - N_v)$ times the multiplication map $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow \mathbb{1}_v$.
- For distinct $v, w \in \mathbf{V}'(D)$, the commutator between degree v and degree $-w$ is 0.

The commutator between degree 0 and degree 1 follows our previous definitions.

- For $i \in \mathbf{E}(D)$, $V_i \otimes V_i^* \subseteq \mathfrak{l}$ acts on $V_{i,t}$ via the usual action of $V_i \otimes V_i^*$ on V_i , and likewise for $V_{i,s}^*$. On all other degree-1 parts, $V_i \otimes V_i^*$ acts as 0.
- For $i \in \mathbf{E}(D)$, $\mathbb{1}_{i,s}$ acts on $V_{i,s}^*$ as 1, $\mathbb{1}_{s(i),+}$ as -1 , and all other degree-1 parts as 0.
- Likewise, $\mathbb{1}_{i,t}$ acts on $V_{i,t}$ as 1, $\mathbb{1}_{t(i),+}$ as -1 , and all other degree-1 parts as 0.
- For $v \in \mathbf{V}(D)$, $\mathbb{1}_v$ acts on $\mathbb{1}_{v,+}$ as 1 and all other degree-1 parts as 0.
- $\mathbb{1}'_v$ is central and hence acts on all degree-1 parts as 0.

The commutator between degree 0 and degree -1 is defined similarly. The full commutator map is determined by the above commutators using anticommutativity and the Jacobi identity.

In this setup, the coroot corresponding to v is

$$(5) \quad \alpha_v^\vee = \mathbb{1}'_v - \left(\sum_{i \in S'_v} \mathbb{1}_{i,s} \right) - \left(\sum_{i \in T'_v} \mathbb{1}_{i,t} \right) + (2 - N_v) \mathbb{1}_v,$$

where $\mathbb{1}'_v \in \mathbb{1}'_v$. The coroots $\alpha_{i,s}^\vee$ and $\alpha_{i,t}^\vee$ remain as before.

Proposition 3.9. *For D and δ as above (with δ_e 's all algebraically independent), $\text{KM}(D, \delta)$ is a Lie algebra object.*

The proof of this is similar to the proof of Proposition 3.5 in [13].

Proof. The commutator map is anticommutative by definition, so it suffices to verify the Jacobi identity (which also shows our definition of commutator is consistent). We will prove that for any graded parts X , Y , and Z of $\text{KM}(D, \delta)$, the map

$$[\bullet, \bullet] \circ (\text{Id} \otimes [\bullet, \bullet]) \circ (\text{Id} + c_{(123)} + c_{(132)}) : X \otimes Y \otimes Z \rightarrow \text{KM}(D, \delta)$$

is identically 0. For fixed X , Y , and Z , this can be expressed as a set of rational identities in δ_e 's for $e \in \mathbf{E}(D)$.

Suppose instead that all δ_e 's are positive integers. Recall we have a symmetric tensor functor from $\boxtimes_{e \in \mathbf{E}(D)} \underline{\text{Rep}}(GL_{\mathbf{n}_{\delta_e}})$ to $\text{Rep}(\prod_{e \in \mathbf{E}(D)} GL_{\mathbf{n}_{\delta_e}})$. This functor maps X , Y , and Z to the corresponding graded parts X' , Y' , and Z' of a central extension of $\mathfrak{g}(D, \delta)$. We know the Jacobi identity is satisfied on $X' \otimes Y' \otimes Z'$, and for large enough δ_e , the functor is fully faithful and surjective at X , Y , and Z (which have fixed degree). Thus, the rational identities as above are satisfied for all sufficiently large integers δ_e .

We conclude that these rational identities all vanish, and the Jacobi identity is satisfied for all δ . $\square$

We now compute the center of $\text{KM}(D, \delta)$.

Proposition 3.10. *The center of $\text{KM}(D, \delta)$ is generated by the images of the following maps:*

- for all $i \in \mathbf{E}(D) \setminus \mathbf{L}(D)$,

$$\text{Id} \oplus \text{Id} \oplus \text{coev}_{V_i} \oplus (-\text{Id}) \oplus (-\text{Id}) : \mathbb{1} \rightarrow \mathbb{1}_{s(i)} \oplus \mathbb{1}_{i,s} \oplus V_i \otimes V_i^* \oplus \mathbb{1}_{i,t} \oplus \mathbb{1}_{t(i)};$$

- for all $i \in \mathbf{L}(D)$,

$$\text{Id} \oplus \text{coev}_{V_i} \oplus (-\text{Id}) : \mathbb{1}_{i,s} \oplus V_i \otimes V_i^* \oplus \mathbb{1}_{i,t};$$

- and for all $v \in \mathbf{V}(D)$,

$$\text{Id} : \mathbb{1} \rightarrow \mathbb{1}'_v.$$

To prove this, we will assume the center is in $\mathfrak{l}$. This is not difficult to show using the invariant bilinear form, which we define in [Section 4.2](#). Any proper ideal (including the center) must be contained in the kernel of this form, and our definition of the form will not depend on δ . For large enough integer δ_e , the form is nondegenerate between $\mathfrak{u}_+$ and $\mathfrak{u}_-$ (as this is true of any Kac-Moody Lie algebra), so for all δ the kernel of the form does not intersect $\mathfrak{u}_+$ or $\mathfrak{u}_-$.

Proof. We first show the images of the above maps are central. Note that $\mathbb{1}'_v$ is central by definition.

We now show that for $i \in \mathbf{E}(D) \setminus \mathbf{L}(D)$, the image Z_i of the map

$$\text{Id} \oplus \text{Id} \oplus \text{coev}_{V_i} \oplus (-\text{Id}) \oplus (-\text{Id}) : \mathbb{1} \rightarrow \mathbb{1}_{s(i)} \oplus \mathbb{1}_{i,s} \oplus V_i \otimes V_i^* \oplus \mathbb{1}_{i,t} \oplus \mathbb{1}_{t(i)}$$

is central. It suffices to show Z_i commutes with $V_i \otimes V_i^*$, $e_{i,s}$, $e_{i,t}$, $e_{s(i)}$, and $e_{t(i)}$ (negative Chevalley generators follow similarly), as it acts trivially on all other generators.

- $\mathbb{1}_{s(i)}$, $\mathbb{1}_{i,s}$, $\mathbb{1}_{i,t}$, and $\mathbb{1}_{t(i)}$ all act trivially on $V_i \otimes V_i^*$, and the image of $\text{coev}_{V_i} : \mathbb{1} \rightarrow V_i \otimes V_i^*$ is central in $V_i \otimes V_i^*$. Thus, Z_i commutes with $V_i \otimes V_i^*$.
- Z_i acts on $e_{i,s}$ as $0 + 1 + (-1) + 0 + 0 = 0$.
- Z_i acts on $e_{i,t}$ as $0 + 0 + 1 + (-1) + 0 = 0$.
- Z_i acts on $e_{s(i)}$ as $1 + (-1) + 0 + 0 + 0 = 0$.
- Z_i acts on $e_{t(i)}$ as $0 + 0 + 0 + 1 + (-1) = 0$.

Thus, Z_i is central. A similar argument shows that for all $i \in \mathbf{L}(D)$, the image Z_i of the map

$$\text{Id} \oplus \text{coev}_{V_i} \oplus (-\text{Id}) : \mathbb{1}_{i,s} \oplus V_i \otimes V_i^* \oplus \mathbb{1}_{i,t}$$

is central; the only difference is that the last two cases above are combined.

It remains to show that these generate the entire center. The center must lie in

$$\mathfrak{z} = \left(\bigoplus_{e \in \mathbf{E}(D)} V_i \otimes V_i^* \right) \oplus \left(\bigoplus_{e \in \mathbf{E}(D)} \mathbb{1}_{i,s} \right) \oplus \left(\bigoplus_{e \in \mathbf{E}(D)} \mathbb{1}_{i,t} \right) \oplus \left(\bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}_v \right) \oplus \left(\bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}'_v \right).$$

Assume for sake of contradiction that the center contains some other part Z not generated by Z_i for $i \in \mathbf{E}(D)$ and $\mathbb{1}'_v$ for $v \in \mathbf{V}(D)$. As the center of $V_i \otimes V_i^*$ is the image of $\text{coev}_{V_i} : \mathbb{1} \rightarrow V_i \otimes V_i^*$, the component of Z in $V_i \otimes V_i^*$ must lie in this image. By factoring through the maps from $\mathbb{1}$ to Z_i , we may assume the component of Z in $V_i \otimes V_i^*$ is 0 for all $i \in \mathbf{E}(D)$. Then, the only component of Z that can act nontrivially on $e_{i,s}$ is $\mathbb{1}_{i,s}$, so the component of Z in $\mathbb{1}_{i,s}$ is also 0, and likewise in $\mathbb{1}_{i,t}$. This implies that for $v \in \mathbf{V}(D)$, the only component of Z that can act nontrivially on e_v is $\mathbb{1}_v$, so the component of Z in $\mathbb{1}_v$ is also 0. This means Z is generated by $\mathbb{1}'_v$ for $v \in \mathbf{V}(D)$, contradiction.

Thus, Z_i for $i \in \mathbf{E}(D)$ and $\mathbb{1}'_v$ for $v \in \mathbf{V}(D)$ together generate the center of $\text{KM}(D, \delta)$. $\square$

4. PROPERTIES OF KAC-MOODY LIE ALGEBRAS IN DELIGNE CATEGORIES

In this section, we generalize some of the properties of classical Kac-Moody Lie Algebras to our construction of $\text{KM}(D, \delta)$.

4.1. Cartan Involution. Let KM_1 be the object of positive-degree generators in the Deligne category, such that the positive component of the Kac-Moody algebra is the quotient $\mathfrak{u}_+ = FL(\text{KM}_1)/I_+$, where I_+ denotes the positive part of the Serre ideal.

$$\text{KM}_1 = \bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}_{v,+} \oplus \bigoplus_{i \in E(D)} V_{i,s}^* \oplus \bigoplus_{i \in E(D)} V_{i,t}.$$

Similarly, let

$$\text{KM}_{-1} = \bigoplus_{v \in \mathbf{V}(D)} \mathbb{1}_{v,-} \oplus \bigoplus_{i \in E(D)} V_{i,s} \oplus \bigoplus_{i \in E(D)} V_{i,t}^*.$$

The degree-0 Cartan-like component incorporates the standard elements, the central elements $\mathbb{1}'_v$, and the trivial objects associated with the intermediate vertices:

$$\text{KM}_0 = \bigoplus_{i \in E(D)} (V_i \otimes V_i^*) \oplus \bigoplus_{v \in \mathbf{V}(D)} (\mathbb{1}_v \oplus \mathbb{1}'_v) \oplus \bigoplus_{i \in E(D)} (\mathbb{1}_{i,s} \oplus \mathbb{1}_{i,t})$$

Let

$$F : \boxtimes_{e \in \mathbf{E}(D)} \underline{\text{Rep}}(GL_{\mathbf{n}_e}) \rightarrow \boxtimes_{e \in \mathbf{E}(D)} \underline{\text{Rep}}(GL_{\mathbf{n}_e})$$

be the symmetric monoidal functor from the product of Deligne categories to itself that swaps V_i and V_i^* for all $i \in E(D)$. We define the target Lie algebra as $\text{KM}^* = F(\text{KM})$. Its bracket is canonically induced by the functor:

$$[\bullet, \bullet]_{\text{KM}^*} = F([\bullet, \bullet]_{\text{KM}}).$$

To abstract the standard Cartan involution, we will define a map $\omega : \text{KM} \rightarrow \text{KM}^*$ that is a Lie algebra homomorphism and satisfies $F(\omega) = \omega^{-1}$.

We define the morphism $\omega : \text{KM} \rightarrow F(\text{KM})$ on the generating components.

- On KM_0 (the Levi subalgebra), ω_0 acts on the $\mathfrak{gl}_i$ components as the negation of the braiding, and on all unit components as the negative identity:

$$\begin{aligned} \omega_0|_{V_i \otimes V_i^*} &= -c_{V_i, V_i^*} : V_i \otimes V_i^* \rightarrow V_i^* \otimes V_i, \\ \omega_0|_{\mathbb{1}_v \oplus \mathbb{1}'_v} &= -\text{Id}_{\mathbb{1}_v} \oplus -\text{Id}_{\mathbb{1}'_v}, \\ \omega_0|_{\mathbb{1}_{i,s} \oplus \mathbb{1}_{i,t}} &= -\text{Id}_{\mathbb{1}_{i,s}} \oplus -\text{Id}_{\mathbb{1}_{i,t}}. \end{aligned}$$

- On KM_1 and KM_{-1} (the generators), $\omega_{\pm 1}$ acts strictly as the negative identity map into the functorial image:

$$\begin{aligned}\omega_1|_{V_{i,s}^*} &= -\text{Id}_{V_i^*} : V_{i,s}^* \rightarrow F(V_{i,s}), \\ \omega_1|_{V_{i,t}} &= -\text{Id}_{V_i} : V_{i,t} \rightarrow F(V_{i,t}^*).\end{aligned}$$

Thus, ω_1 maps KM_1 into $F(\text{KM}_{-1})$, and in fact negates the $\Gamma(D)$ -grading of KM . The map ω_{-1} is defined symmetrically.

Proposition 4.1. *The map ω extends to a well-defined homomorphism of Lie algebras from KM to $(\text{KM})^*$.*

Proof. To establish that ω is a Lie algebra homomorphism, we must first show that it commutes with the brackets on the generating components, i.e.,

$$\omega \circ [\bullet, \bullet] = [\bullet, \bullet] \circ (\omega \otimes \omega).$$

- **Case 1: the bracket between $V_{i,t}$ and $V_{i,t}^*$**

The bracket between $V_{i,t}$ and $V_{i,t}^*$ in KM is given by

$$\text{Id} \oplus \text{ev}_{V_i^*} : V_{i,t} \otimes V_{i,t}^* \rightarrow (V_i \otimes V_i^*) \oplus \mathbb{1}_{i,t}.$$

Thus, $\omega \circ [\bullet, \bullet]$ is given by

$$(-c_{V_i, V_i^*}) \oplus (\text{ev}_{V_i^*}) : V_{i,t} \otimes V_{i,t}^* \rightarrow F(V_i \otimes V_i^*) \oplus F(\mathbb{1}_{i,t}).$$

To compute $[\bullet, \bullet] \circ (\omega \otimes \omega)$, recall that ω maps $V_{i,t} \rightarrow F(V_{i,t}^*)$ and $V_{i,t}^* \rightarrow F(V_{i,t})$. Because both maps are $-\text{Id}$, their tensor product is Id . The bracket between $F(V_{i,t}^*)$ and $F(V_{i,t})$ in $F(\text{KM})$ is given by

$$(-c_{V_i, V_i^*}) \oplus (-\text{ev}_{V_i^*}) : F(V_{i,t}^*) \otimes F(V_{i,t}) \rightarrow F(V_i \otimes V_i^*) \oplus F(\mathbb{1}_{i,t}).$$

Thus, $[\bullet, \bullet] \circ (\omega \otimes \omega)$ is also given by

$$(-c_{V_i, V_i^*}) \oplus (-\text{ev}_{V_i^*}) : V_{i,t} \otimes V_{i,t}^* \rightarrow F(V_i \otimes V_i^*) \oplus F(\mathbb{1}_{i,t}).$$

Hence, on $V_{i,t} \otimes V_{i,t}^*$, we have $\omega \circ [\bullet, \bullet] = [\bullet, \bullet] \circ (\omega \otimes \omega)$.

- **Case 2: the bracket between $V_{i,s}^*$ and $V_{i,s}$**

The bracket between $V_{i,s}^*$ and $V_{i,s}$ in KM is given by

$$\text{ev}_{V_i} \oplus (-c_{V_i^*, V_i}) : V_{i,s}^* \otimes V_{i,s} \rightarrow \mathbb{1}_{i,s} \oplus (V_i \otimes V_i^*).$$

Thus, $\omega \circ [\bullet, \bullet]$ is given by

$$(-\text{ev}_{V_i}) \oplus \text{Id}_{V_i^* \otimes V_i} : V_{i,s}^* \otimes V_{i,s} \rightarrow F(\mathbb{1}_{i,s}) \oplus F(V_i \otimes V_i^*).$$

To compute $[\bullet, \bullet] \circ (\omega \otimes \omega)$, recall that ω maps $V_{i,s}^* \rightarrow F(V_{i,s})$ and $V_{i,s} \rightarrow F(V_{i,s}^*)$. Because both maps are $-\text{Id}$, their tensor product is Id . The bracket between $F(V_{i,s})$ and $F(V_{i,s}^*)$ in $F(\text{KM})$ is given by

$$(-\text{ev}_{V_i}) \oplus \text{Id}_{V_i^* \otimes V_i} : F(V_{i,s}) \otimes F(V_{i,s}^*) \rightarrow F(\mathbb{1}_{i,s}) \oplus F(V_i \otimes V_i^*).$$

Thus, $[\bullet, \bullet] \circ (\omega \otimes \omega)$ is also given by

$$(-\text{ev}_{V_i}) \oplus \text{Id}_{V_i^* \otimes V_i} : V_{i,s}^* \otimes V_{i,s} \rightarrow F(\mathbb{1}_{i,s}) \oplus F(V_i \otimes V_i^*).$$

Hence, on $V_{i,s}^* \otimes V_{i,s}$, we have $\omega \circ [\bullet, \bullet] = [\bullet, \bullet] \circ (\omega \otimes \omega)$.

- **Case 3: the bracket between $\mathbb{1}_{v,+}$ and $\mathbb{1}_{v,-}$**

The bracket between $\mathbb{1}_{v,+}$ and $\mathbb{1}_{v,-}$ in KM is given by a direct sum of scalar multiples of multiplication maps $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow \mathbb{1}_0$ where $\mathbb{1}_0$ ranges over $\mathbb{1}_v, \mathbb{1}'_v, \mathbb{1}_{i,s}$ for $i \in S'_v$, and $\mathbb{1}_{i,t}$ for $i \in T'_v$. Consider any one such component $\mathbb{1}_0$, and suppose the component of the bracket in $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow \mathbb{1}_0$ is given by c times the multiplication map for a scalar c . As ω

is $-\text{Id}$ on $\mathbb{1}_0$, we conclude the component of $\omega \circ [\bullet, \bullet]$ in $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow F(\mathbb{1}_0)$ is $-c$ times the multiplication map.

To compute $[\bullet, \bullet] \circ (\omega \otimes \omega)$, recall that ω maps $\mathbb{1}_{v,+} \rightarrow F(\mathbb{1}_{v,-})$ and $\mathbb{1}_{v,-} \rightarrow F(\mathbb{1}_{v,+})$. Because both maps are $-\text{Id}$, their tensor product is Id . Consider the bracket between $F(\mathbb{1}_{v,-})$ and $F(\mathbb{1}_{v,+})$ in $F(\text{KM})$. Its component in $F(\mathbb{1}_0)$ is given by $-c$ times the multiplication map. Thus, $[\bullet, \bullet] \circ (\omega \otimes \omega)$ is also given by $-c$ times the multiplication map.

Hence, on $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-}$, we have $\omega \circ [\bullet, \bullet] = [\bullet, \bullet] \circ (\omega \otimes \omega)$.

• **Case 4: the remaining brackets between degree 1 and degree -1**

All such brackets are 0 in KM , as the bracket between degree v and degree $-w$ is 0 when $v, w \in \Gamma(D)$ with $v \neq w$. As ω negates degrees, the dual bracket after applying $\omega \otimes \omega$ is between degree $-v$ and degree w , and hence also 0. Thus, in this case, $\omega \circ [\bullet, \bullet] = [\bullet, \bullet] \circ (\omega \otimes \omega) = 0$.

Thus, ω commutes with the bracket between all of KM_1 and all of KM_{-1} . By the universal property of the free Lie algebra, the mapping on generators uniquely extends to a homomorphism on the entire free Lie algebra:

$$\tilde{\omega} : FL(\text{KM}_1) \rightarrow FL(F(\text{KM}_1)) \cong F(FL(\text{KM}_1))$$

Because $FL_k(\text{KM}_1) \subset \text{KM}_1^{\otimes k}$, the action on the k -th graded component is precisely

$$\tilde{\omega}_k = F\left((- \text{Id}_{\text{KM}_1})^{\otimes k}\right) = (-1)^k F(\text{Id}_{\text{KM}_1}^{\otimes k})$$

To prove $\tilde{\omega}$ descends to a valid homomorphism $\omega : \text{KM} \rightarrow F(\text{KM})$, we must show it preserves the Serre ideal, meaning $\tilde{\omega}(I_+) \subseteq F(I_+)$. By the established propositions, I_+ is entirely generated by its degree-2 and degree-3 components.

Degree 2 Relations

For degree 2, we have $k = 2$, meaning $\tilde{\omega}_2 = (-1)^2 F = +F$. We check this against the degree 2 relations matching the numbering in Proposition 3.4:

- (1) $\Lambda^2 V_{i,d}^{(*)}$: The map $\tilde{\omega}_2$ sends this to $F(\Lambda^2 V_{i,d}^{(*)})$. This exactly matches relation (1) for the dual. Therefore, it is contained in $F(I_+)$.
- (2) $\mathbb{1}_{v,+} \otimes \mathbb{1}_{w,+}$: Applying $\tilde{\omega}_2$ yields $F(\mathbb{1}_{v,+} \otimes \mathbb{1}_{w,+})$. This maps to the dual of relation (2). Hence, it is in $F(I_+)$.
- (3) $V_{i,d_1}^{(*)} \otimes V_{j,d_2}^{(*)}$: The map $\tilde{\omega}_2$ sends this to $F(V_{i,d_1}^{(*)} \otimes V_{j,d_2}^{(*)})$. This matches relation (3) for the dual. Thus, it is in $F(I_+)$.
- (4) $\mathbb{1}_{v,+} \otimes V_{i,d}^{(*)}$ (for $v \neq d(i)$): Applying $\tilde{\omega}_2$ yields $F(\mathbb{1}_{v,+} \otimes V_{i,d}^{(*)})$. This explicitly matches relation (4) for the dual. Hence, it is in $F(I_+)$.
- (5) $L_{[(1),(1)]} V_i$: The map $\tilde{\omega}_2$ evaluates to $F(L_{i,+,[(1),(1)]})$. This matches the dual of relation (5). Therefore, it falls in $F(I_+)$.

Degree 3 Relations

- (6) $S^2 \mathbb{1}_{d(i),+} \otimes V_{i,d}^{(*)}$: The mapping $\tilde{\omega}_3$ sends this to $-F(S^2 \mathbb{1}_{d(i),+} \otimes V_{i,d}^{(*)})$. This equals -1 times relation (6) for the dual. Because $F(I_+)$ is an ideal, it absorbs the scalar -1 . Therefore, it is in $F(I_+)$.
- (7) $S^2 V_{i,d}^{(*)} \otimes \mathbb{1}_{d(i),+}$: Applying $\tilde{\omega}_3$ evaluates to $-F(S^2 V_{i,d}^{(*)} \otimes \mathbb{1}_{d(i),+})$. This results in -1 times relation (7) for the dual. Since $F(I_+)$ is an ideal, it absorbs the scalar. Hence, it is contained in $F(I_+)$.

Every generator of I_+ explicitly maps into $F(I_+)$. Therefore, ω factors through the quotient. We must check that $F(\omega) \circ \omega = \text{Id}_{\text{KM}}$ holds identically across all components.

Evaluating the composition component by component:

- On $\mathfrak{gl}_i \subset \text{KM}_0$: $\omega_0 = -c_{V_i, V_i^*}$.

$$F(\omega_0) \circ \omega_0 = F(-c_{V_i, V_i^*}) \circ (-c_{V_i, V_i^*}) = (-c_{V_i^*, V_i}) \circ (-c_{V_i, V_i^*}) = \text{Id}_{V_i \otimes V_i^*}$$

- On Unit Objects in KM_0 : $\omega_0 = -\text{Id}$.

$$F(-\text{Id}) \circ (-\text{Id}) = (-\text{Id}) \circ (-\text{Id}) = \text{Id}$$

- On $V_i, V_i^* \subset \text{KM}_{\pm 1}$: $\omega_{\pm 1} = -\text{Id}$.

$$F(-\text{Id}) \circ (-\text{Id}) = (-\text{Id}) \circ (-\text{Id}) = \text{Id}$$

- On $\mathbb{1}_{v,+} \subset \text{KM}_1$: $\omega_1 = +\text{Id}$.

$$F(+\text{Id}) \circ (+\text{Id}) = (+\text{Id}) \circ (+\text{Id}) = \text{Id}$$

- On $\mathbb{1}_{v,-} \subset \text{KM}_{-1}$: $\omega_{-1} = -\text{Id}$.

$$F(-\text{Id}) \circ (-\text{Id}) = (-\text{Id}) \circ (-\text{Id}) = \text{Id}$$

Because $F(\omega) \circ \omega$ acts as the identity morphism on every generating subobject, $\omega^{-1} = F(\omega)$ is satisfied globally. $\square$

4.2. Invariant Bilinear Form. We seek to generalize the invariant bilinear form of the Kac-Moody algebra. To do this, we will assign this form in a consistent way on the generators and extend through invariance. As in the classical case, the pairing between degrees γ_1 and $\gamma_2 \in \Gamma(D)$ is 0 unless $\gamma_1 = -\gamma_2$.

We first define the symmetric bilinear form $(\bullet, \bullet)$ on the Levi component $\mathfrak{l}$. On $\mathfrak{gl}_{\delta_i} = V_i \otimes V_i^*$, we define the pairing to be the trace form given by the composition

$$(V_i \otimes V_i^*) \otimes (V_i \otimes V_i^*) \xrightarrow{c(1234)} V_i^* \otimes V_i \otimes V_i^* \otimes V_i \xrightarrow{\text{ev} \otimes \text{ev}} \mathbb{1} \otimes \mathbb{1} \rightarrow \mathbb{1}.$$

(In the integer case, this is simply the trace form $(A, B) = \text{tr} AB$.) For $i \in \mathbf{E}(D)$, we define

$$(1_{i,s}, 1_{i,s}) = (1_{i,t}, 1_{i,t}) = 1.$$

(More precisely, the pairings on $\mathbb{1}_{i,s} \otimes \mathbb{1}_{i,s}$ and $\mathbb{1}_{i,t} \otimes \mathbb{1}_{i,t}$ are both given by 1 times the multiplication map.)

For $v \in \mathbf{V}(D)$, if $N_v \neq 2$, we define

$$(1'_v, 1'_v) = 0 \quad \text{and} \quad (1_v, 1_v) = \frac{1}{2 - N_v}.$$

If $N_v = 2$, then we define

$$(1'_v, 1'_v) = (1_v, 1_v) = 0 \quad \text{and} \quad (1_v, 1'_v) = (1'_v, 1_v) = 1.$$

Elements from distinct summands of $\mathfrak{l}$ will pair as 0 (except for $(1_v, 1'_v) = (1'_v, 1_v)$ when $N_v = 2$).

Remark 4.2. The reason why $N_v = 2$ is handled separately is the same as why $\mathbb{1}'_v$ was introduced in the first place: to handle affine-like behavior. In the case $N_v \neq 2$, the element $1'_v$ is in the kernel of the pairing, as we would expect from a trivial central extension.

There are two cases where $N_v = 2$: either D is a single vertex v with a single self-loop, or v is incident to two distinct edges. In the former case, we are interpolating the affine Lie algebra $\hat{A}_n$, and our pairing on 1_v and $1'_v$ mirrors the pairing on the central element K and the grading element d in $\hat{A}_n$, where $(K, K) = (d, d) = 0$ and $(K, d) = 1$.

In the latter case, suppose v is incident to two distinct edges i and j . These two edges correspond to copies of $\mathfrak{gl}_{\delta_i+2}$ and $\mathfrak{gl}_{\delta_j+2}$. The subalgebra generated by these along with e_v and f_v almost form

a copy of $\mathfrak{gl}_{\delta_i+\delta_j+4}$, described as follows:

$$\begin{pmatrix} \mathbb{1}_{i,s} & V_{i,s}^* & \mathbb{1}_{i,+} & & & \\ V_{i,s} & \mathfrak{gl}_{\delta_i+2} & V_{i,t} & & & \\ \mathbb{1}_{i,-} & V_{i,t}^* & \mathbb{1}_{i,t} & \mathbb{1}_{v,+} & & \\ & & \mathbb{1}_{v,-} & \mathbb{1}_{j,s} & V_{j,s}^* & \mathbb{1}_{j,+} \\ & & & V_{j,s} & \mathfrak{gl}_{\delta_j+2} & V_{j,t} \\ & & & \mathbb{1}_{j,-} & V_{j,t}^* & \mathbb{1}_{j,t} \end{pmatrix}.$$

However, $\mathbb{1}_v$ cannot be realized as an element of $\mathfrak{gl}_{\delta_i+\delta_j+4}$, leading to affine-like behavior.

Remark 4.3. In the classical case of a simply-laced Dynkin diagram, every coroot has squared length 2, and adjacent coroots pair as -1 . We can check that our definition of the pairing also satisfies this in the case where δ is integer-valued.

Notice that for $v \in \mathbf{V}(D)$, we have

$$\begin{aligned} (\alpha_v^\vee, \alpha_v^\vee) &= \left(- \left(\sum_{i \in S'_v} 1_{i,s} \right) - \left(\sum_{i \in T'_v} 1_{i,t} \right), - \left(\sum_{i \in S'_v} 1_{i,s} \right) - \left(\sum_{i \in T'_v} 1_{i,t} \right) \right) + (2 - N_v)^2 (1_v, 1_v) \\ &= N_v + (2 - N_v)^2 \frac{1}{2 - N_v} \\ &= 2. \end{aligned}$$

(If $2 - N_v = 0$, then $(2 - N_v)^2 (1_v, 1_v) = 0 = 2 - N_v$ regardless of $(1_v, 1_v)$.) Also, for $i \in \mathbf{E}(D)$,

$$(\alpha_{i,s}^\vee, \alpha_{i,s}^\vee) = (1_{i,s} - E_{1,1}^i, 1_{i,s} - E_{1,1}^i) = (1_{i,s}, 1_{i,s}) + (E_{1,1}^i, E_{1,1}^i) = 2$$

and similarly $(\alpha_{i,t}^\vee, \alpha_{i,t}^\vee) = 2$, so all coroots have squared length 2.

For adjacent coroots, observe that for $i \in \mathbf{E}(D)$,

$$(\alpha_{i,s}^\vee, \alpha_{s(i)}^\vee) = (1_{i,s}, -1_{i,s}) = -1 \quad \text{and} \quad (\alpha_{i,t}^\vee, \alpha_{t(i)}^\vee) = (1_{i,t}, -1_{i,t}) = -1.$$

We also check how $\alpha_{i,s}^\vee$ pairs with the coroot of the first intermediate vertex along i :

$$(\alpha_{i,s}^\vee, E_{1,1}^i - E_{2,2}^i) = (-E_{1,1}^i, E_{1,1}^i) = -1,$$

and likewise with $\alpha_{i,t}^\vee$. Thus, all adjacent coroots pair as -1 .

Coroots corresponding to vertices which are not adjacent always pair to 0, as they share no terms. Thus, our definition of the pairing matches the behavior we would get in the classical case.

Next, we define the form between degree 1 and degree -1 . For $i \in \mathbf{E}(D)$, the pairing between degree $v_{i,s}$ and degree $-v_{i,s}$ is given by the evaluation map

$$\text{ev}_{V_i} : V_{i,s}^* \otimes V_{i,s} \rightarrow \mathbb{1},$$

and likewise between degree $v_{i,t}$ and degree $-v_{i,t}$. For $v \in \mathbf{V}(D)$, the pairing between degree v and degree $-v$ is simply the multiplication map $\mathbb{1}_{v,+} \otimes \mathbb{1}_{v,-} \rightarrow \mathbb{1}$. The pairing between degree v and degree $-w$ is 0 if $v \neq w$.

Proposition 4.4. *The above definitions of the pairing on degree 0 and between degrees 1 and -1 extend to a pairing on all of $\text{KM}(D, \delta)$.*

Proof. We first check that the pairing between degrees 1 and -1 is consistent with the pairing on degree 0 by invariance. For this, we will assume δ is integer-valued for ease of notation, but the same arguments work in general.

For $i \in \mathbf{E}(D)$, we first check $(1_{i,s}, [e_{i,s}, f_{i,s}]) = ([1_{i,s}, e_{i,s}], f_{i,s}) = 1$. Indeed,

$$(1_{i,s}, [e_{i,s}, f_{i,s}]) = (1_{i,s}, 1_{i,s} - E_{1,1}^i) = 1.$$

A similar argument shows $(1_{i,t}, [e_{i,t}, f_{i,t}]) = ([1_{i,t}, e_{i,t}], f_{i,t}) = 1$.

We must also check $(-E_{1,1}^i, [e_{i,s}, f_{i,s}]) = ([-E_{1,1}^i, e_{i,s}], f_{i,s}) = 1$. Indeed,

$$(-E_{1,1}^i, [e_{i,s}, f_{i,s}]) = (-E_{1,1}^i, 1_{i,s} - E_{1,1}^i) = 1.$$

A similar argument shows $(E_{\delta_i, \delta_i}^i, [e_{i,t}, f_{i,t}]) = ([E_{\delta_i, \delta_i}^i, e_{i,t}], f_{i,t}) = 1$.

For $v \in \mathbf{V}(D)$, we check $(1_v, [e_v, f_v]) = ([1_v, e_v], f_v) = 1$. If $N_v \neq 2$, then

$$(1_v, [e_v, f_v]) = \left(1_v, 1'_v - \sum_{i \in S'_v} 1_{i,s} - \sum_{i \in T'_v} 1_{i,t} + (2 - N_v)1_v \right) = (2 - N_v)(1_v, 1_v) = 1.$$

If $N_v = 2$, then

$$(1_v, [e_v, f_v]) = \left(1_v, 1'_v - \sum_{i \in S'_v} 1_{i,s} - \sum_{i \in T'_v} 1_{i,t} + (2 - N_v)1_v \right) = (1_v, 1'_v) = 1.$$

We must also check that $(1'_v, [e_v, f_v]) = ([1'_v, e_v], f_v) = 0$. Indeed,

$$(1'_v, [e_v, f_v]) = \left(1'_v, 1'_v - \sum_{i \in S'_v} 1_{i,s} - \sum_{i \in T'_v} 1_{i,t} + (2 - N_v)1_v \right) = (2 - N_v)(1'_v, 1_v).$$

As $(1'_v, 1_v) = 0$ when $N_v \neq 2$, this quantity is always 0.

We must further check that for $i \in S'_v$, $(-1_{i,s}, [e_v, f_v]) = ([-1_{i,s}, e_v], f_v) = 1$. Indeed,

$$(-1_{i,s}, [e_v, f_v]) = \left(-1_{i,s}, 1'_v - \sum_{i \in S'_v} 1_{i,s} - \sum_{i \in T'_v} 1_{i,t} + (2 - N_v)1_v \right) = 1.$$

A similar argument shows that for $i \in T'_v$, $(-1_{i,t}, [e_v, f_v]) = ([-1_{i,t}, e_v], f_v) = 1$.

The remaining identities for invariance between degrees 1 and -1 are trivial. For example, for $i \in \mathbf{E}(D)$ and $v \in \mathbf{V}(D)$, we have $(1_v, [e_{i,s}, f_{i,s}]) = 0$ because $[e_{i,s}, f_{i,s}] = 1_{i,s} - E_{1,1}^i$ has no 1_v term, and $([1_v, e_{i,s}], f_{i,s}) = 0$ because 1_v acts trivially on $e_{i,s}$. The same argument holds for the other remaining identities.

Thus, the pairing between degrees 1 and -1 is consistent with the pairing on degree 0.

The proof that the pairing extends consistently to all of $\mathbf{KM}(D, \delta)$ is similar to the proof of [Proposition 3.9](#). At any fixed degrees, the pairing extending can be written as a set of rational identities in δ_e 's. These identities are satisfied for all sufficiently large integer δ_e , as $\mathfrak{g}(D, \delta)$ is a central extension of a Kac-Moody Lie algebra. Thus, they are always satisfied. $\square$

4.3. Casimir Operator. First suppose $\mathfrak{g}(A) = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$ is a Kac-Moody Lie algebra. If V is a $\mathfrak{g}(A)$ -module, let $P(V)$ be the set of nonzero weights of V , and let $D(\lambda) = \{\mu \in \mathfrak{h}^* | \lambda - \mu \in Q_+\}$.

Let $\mathcal{O}$ be the Bernstein-Gelfand-Gelfand category, which is the full subcategory of $\mathfrak{g}(A)$ -mod of modules that are $\mathfrak{h}$ -diagonalizable with finite-dimensional weight spaces such that there exist $\lambda_1, \dots, \lambda_n \in \mathfrak{h}^*$ so that $P(V) \subset \bigcup_{i=1}^n D(\lambda_i)$. Then, there exists a [Casimir operator](#) Ω which is a natural transformation from the identity functor of $\mathcal{O}$ to itself. In other words, for each object $V \in \text{Ob}(\mathcal{O})$, we have an endomorphism $\Omega_V : V \rightarrow V$ such that the following diagram commutes for all morphisms $f : V \rightarrow W$ in $\mathcal{O}$: For any $\mathfrak{g}(A)$ -module V , the universal enveloping algebra $\mathcal{U}(\mathfrak{g}(A))$

$$\begin{array}{ccc}
V & \xrightarrow{f} & W \\
\downarrow \Omega_V & & \downarrow \Omega_W \\
V & \xrightarrow{f} & W
\end{array}$$

FIGURE 3. The commuting property of the Casimir operator.

also acts on V . Let $\mathcal{U}_{\leq 2}(\mathfrak{g}) \subseteq \mathcal{U}(\mathfrak{g})$ denote the elements of degree at most 2. As constructed in [11], Ω is a sum of elements in $\mathcal{U}_{\leq 2}(\mathfrak{g})$ (or rather its appropriate completion), and for any $V \in \text{Ob}(\mathcal{O})$, only finitely many of the root spaces act as nonzero operators (that is, objects in $\mathcal{O}$ are restricted representations). Specifically, let $\rho \in \mathfrak{h}$ such that $\langle \rho, \alpha_i \rangle = 1$ for each simple root α_i . Let $\{u_1, u_2, \dots\}$ and $\{u^1, u^2, \dots\}$ be dual bases of $\mathfrak{h}$ under the invariant bilinear form restricted to $\mathfrak{h} \times \mathfrak{h}$, and for each positive root $\alpha \in \Delta_+$, let $\{e_{-\alpha}^1, e_{-\alpha}^2, \dots\}$ and $\{e_{\alpha}^1, e_{\alpha}^2, \dots\}$ be dual bases under the form restricted to $\mathfrak{g}_{-\alpha} \times \mathfrak{g}_{\alpha}$. Then,

$$(6) \quad \Omega = 2\rho + \sum_j u^j u_j + 2 \sum_{\alpha \in \Delta_+} \sum_j e_{-\alpha}^j e_{\alpha}^j.$$

One important property is that the Casimir element acts as a scalar on any highest weight $\mathfrak{g}(A)$ -module.

We will generalize this construction of Ω to our setting $\mathfrak{g} = \mathfrak{g}(D, \mathbf{N})$. Note that for any Lie algebra object L in a symmetric tensor category, we have a well-defined notion of $\mathcal{U}_{\leq 2}(L)$, i.e., the cokernel of the map

$$(\text{Id} - c_{L,L}) \oplus (-[\bullet, \bullet]) \oplus 0 : L \otimes L \rightarrow (L \otimes L) \oplus L \oplus \mathbb{1}.$$

Furthermore, $\mathcal{U}_{\leq 2}(L)$ acts on any L -module. Our goal is thus to define Ω as a sum of maps $\mathbb{1} \rightarrow \mathcal{U}_{\leq 2}(\mathfrak{g})$, which induces an automorphism on any $V \in \mathcal{O}$ given by the composition

$$V \rightarrow \mathbb{1} \otimes V \xrightarrow{\Omega \otimes \text{Id}} \mathcal{U}_{\leq 2}(\mathfrak{g}) \otimes V \xrightarrow{\text{act}} V.$$

More precisely, as in [11], Ω will be given by a sum of elements in either $\mathfrak{h} \cdot \mathfrak{h}$ or $\mathfrak{g}_{-\alpha} \cdot \mathfrak{g}_{\alpha}$ for positive roots α . Only finitely many of these terms can be nonzero on V , so this composition is well-defined.

First assume $\mathbf{N}$ is integer-valued. Recall that for $\mathfrak{gl}_n$,

$$\rho = \sum_{j=1}^n \frac{n+1-2j}{2} E_{j,j} = \begin{pmatrix} \frac{n-1}{2} & 0 & \dots & 0 & 0 \\ 0 & \frac{n-3}{2} & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -\frac{n-3}{2} & 0 \\ 0 & 0 & \dots & 0 & -\frac{n-1}{2} \end{pmatrix},$$

the unique trace-0 diagonal matrix with consecutive entries differing by 1. Also, under the usual trace form on $\mathfrak{gl}_n$, the elementary matrices $\{E_{i,j}\}_{i,j}$ form a basis dual to itself, with $E_{i,j}$ dual to $E_{j,i}$. Thus, following Equation (6), we define

$$\Omega_0 = \sum_{i \in \mathbf{E}(D)} \sum_{j=1}^{n_i} (n+1-2j) E_{j,j}^i + \sum_{i \in \mathbf{E}(D)} \sum_{j=1}^{n_i} (E_{j,j}^i)^2 + 2 \sum_{i \in \mathbf{E}(D)} \sum_{1 \leq j < k \leq n_i} E_{k,j}^i E_{j,k}^i + 2 \sum_{\alpha \in \Delta'_+} \sum_j e_{-\alpha}^j e_{\alpha}^j,$$

where Δ'_+ consists of all roots in Δ_+ except those which are roots of $\mathfrak{gl}_{n_i}$ for a single edge $i \in \mathbf{E}(D)$ (as these are already accounted for in the $E_{k,j}^i E_{j,k}^i$ terms).

By Theorem 2.6 of [11], Ω_0 commutes with each $\mathfrak{gl}_{n_i}$. We will determine the necessary adjustments to make it commute with the remaining generators e_v , $e_{i,s}$, $e_{i,t}$, and their negative-degree counterparts. Note that the $E_{k,j}^i E_{j,k}^i$ terms are necessary; even though $E_{j,k}^i$ has degree 0 in our setup, its degree is positive in the setup of [11].

We can simplify the above expression. Note that for any edge i ,

$$\begin{aligned}
\sum_{j=1}^{n_i} \sum_{k=1}^{n_i} E_{k,j}^i E_{j,k}^i &= \sum_{j=1}^{n_i} (E_{j,j}^i)^2 + \sum_{1 \leq j < k \leq n_i} (E_{k,j}^i E_{j,k}^i + E_{j,k}^i E_{k,j}^i) \\
&= \sum_{j=1}^{n_i} (E_{j,j}^i)^2 + \sum_{1 \leq j < k \leq n_i} (2E_{k,j}^i E_{j,k}^i + [E_{j,k}^i, E_{k,j}^i]) \\
&= \sum_{j=1}^{n_i} (E_{j,j}^i)^2 + 2 \sum_{1 \leq j < k \leq n_i} E_{k,j}^i E_{j,k}^i + \sum_{1 \leq j < k \leq n_i} (E_{j,j}^i - E_{k,k}^i) \\
&= \sum_{j=1}^{n_i} (E_{j,j}^i)^2 + 2 \sum_{1 \leq j < k \leq n_i} E_{k,j}^i E_{j,k}^i + \sum_{j=1}^{n_i} (n+1-2j) E_{j,j}^i,
\end{aligned}$$

so we can rewrite our expression for Ω_0 as

$$(7) \quad \Omega_0 = \sum_{i \in \mathbf{E}(D)} \sum_{j=1}^{n_i} \sum_{k=1}^{n_i} E_{k,j}^i E_{j,k}^i + 2 \sum_{\alpha \in \Delta'_+} \sum_j e_{-\alpha}^j e_{\alpha}^j.$$

For $i \in \mathbf{E}(D)$ and $1 \leq j \leq n_i$, define the elements

$$E_{0,j}^i = [e_{i,s}, E_{1,j}] \in V_{i,s}^* \quad \text{and} \quad E_{j,0}^i = [E_{j,1}, f_{i,s}] \in V_{i,s}.$$

These elements can be thought of as elementary matrices of $\mathfrak{gl}_{n_i+2}$ with rows and columns indexed 0 through $n_i + 1$ (under this notation, $e_{i,s} = E_{0,1}^i$ and $1_{i,s} = E_{0,0}^i$). In particular, $\{E_{1,0}^i, E_{2,0}^i, \dots\}$ and $\{E_{0,1}^i, E_{0,2}^i, \dots\}$ are dual bases of $\mathfrak{g}(D, \mathbf{N})_{-v_{i,s}}$ and $\mathfrak{g}(D, \mathbf{N})_{v_{i,s}}$. Note that

$$[E_{j,0}^i, e_{i,s}] = E_{j,1}^i - \delta_{j,1} 1_{i,s}.$$

By Lemma 2.4 of [11],

$$\begin{aligned}
\left[2 \sum_{\alpha \in \Delta'_+} \sum_j e_{-\alpha}^j e_{\alpha}^j, e_{i,s} \right] &= 2 \sum_{j=1}^{n_i} [E_{j,0}^i, e_{i,s}] E_{0,j}^i \\
&= 2 \sum_{j=1}^{n_i} E_{j,1}^i E_{0,j}^i - 2 \cdot 1_{i,s} E_{0,1}^i \\
&= 2 \sum_{j=1}^{n_i} E_{j,1}^i E_{0,j}^i - 2 \cdot 1_{i,s} e_{i,s}.
\end{aligned}$$

Also, $[E_{j,k}^{i'}, e_{i,s}]$ is nonzero only when $i' = i$ and $j = 1$, so

$$\begin{aligned}
\left[\sum_{i' \in \mathbf{E}(D)} \sum_{j=1}^{n_{i'}} \sum_{k=1}^{n_{i'}} E_{k,j}^{i'} E_{j,k}^{i'}, e_{i,s} \right] &= \sum_{j=1}^{n_i} [E_{1,j}^i, e_{i,s}] E_{j,1}^i + \sum_{k=1}^{n_i} E_{k,1}^i [E_{1,k}, e_{i,s}] \\
&= \sum_{j=1}^{n_i} (-E_{0,j}^i E_{j,1}^i - E_{j,1}^i E_{0,j}^i).
\end{aligned}$$

Thus,

$$\begin{aligned}
[\Omega_0, e_{i,s}] &= \sum_{j=1}^{n_i} (-E_{0,j}^i E_{j,1} - E_{j,1}^i E_{0,j}^i) + 2 \sum_{j=1}^{n_i} E_{j,1}^i E_{0,j}^i - 2 \cdot 1_{i,s} e_{i,s} \\
&= \sum_{j=1}^{n_i} (E_{j,1}^i E_{0,j}^i - E_{0,j}^i E_{j,1}^i) - 2 \cdot 1_{i,s} e_{i,s} \\
&= \sum_{j=1}^{n_i} [E_{j,1}^i, E_{0,j}^i] - 2 \cdot 1_{i,s} e_{i,s} \\
&= -n_i e_{i,s} - 2 \cdot 1_{i,s} e_{i,s}.
\end{aligned}$$

Observe that

$$[1_{i,s}^2, e_{i,s}] = 1_{i,s} e_{i,s} + e_{i,s} 1_{i,s} = 2 \cdot 1_{i,s} e_{i,s} - [1_{i,s}, e_{i,s}] = 2 \cdot 1_{i,s} e_{i,s} - e_{i,s},$$

so defining

$$\Omega_{i,s} = 1_{i,s}^2 + (n_i + 1) 1_{i,s},$$

we have

$$[\Omega_{i,s}, e_{i,s}] = 2 \cdot 1_{i,s} e_{i,s} + n_i e_{i,s} = -[\Omega_0, e_{i,s}].$$

It is also clear that $[\Omega_{i,s}, e_{i,t}] = 0$, and $[\Omega_{i,s}, e_{j,s}] = [\Omega_{i,s}, e_{j,t}] = 0$ for any edge $j \neq i$, as $1_{i,s}$ acts as 0 on these generators. Likewise, $\Omega_{i,s}$ commutes with each $\mathfrak{gl}_{n_i}$.

We similarly compute $[\Omega_0, e_{i,t}]$ to determine the necessary adjustment. Define elements

$$E_{j,n_i+1}^i = [E_{j,n_i}^i, e_{i,t}] \in V_{i,t} \quad \text{and} \quad E_{n_i+1,j}^i = [f_{i,t}, E_{n_i,j}^i] \in V_{i,t}^*.$$

(Under this notation, $e_{i,t} = E_{n_i,n_i+1}^i$ and $1_{i,t} = -E_{n_i+1,n_i+1}^i$.) These are dual bases, so by Lemma 2.4 of [11],

$$\left[2 \sum_{\alpha \in \Delta'_+} \sum_j e_{-\alpha}^j e_{\alpha}^j, e_{i,t} \right] = 2 \sum_{j=1}^{n_i} [E_{n_i+1,j}^i, e_{i,t}] E_{j,n_i+1}^i = -2 \sum_{j=1}^{n_i} E_{n_i,j}^i E_{j,n_i+1}^i - 2 \cdot 1_{i,t} e_{i,t}.$$

Also, $[E_{j,k}^{i'}, e_{i,t}]$ is nonzero only when $i' = i$ and $j = n_i$, so

$$\left[\sum_{i' \in \mathbf{E}(D)} \sum_{j=1}^{n_{i'}} \sum_{k=1}^{n_{i'}} E_{k,j}^{i'} E_{j,k}^{i'}, e_{i,t} \right] = \sum_{j=1}^{n_i} (E_{j,n_i+1}^i E_{n_i,j}^i + E_{n_i,j}^i E_{j,n_i+1}^i).$$

Thus,

$$\begin{aligned}
[\Omega_0, e_{i,t}] &= \sum_{j=1}^{n_i} (E_{j,n_i+1}^i E_{n_i,j}^i + E_{n_i,j}^i E_{j,n_i+1}^i) - 2 \sum_{j=1}^{n_i} E_{n_i,j}^i E_{j,n_i+1}^i - 2 \cdot 1_{i,t} e_{i,t} \\
&= \sum_{j=1}^{n_i} [E_{j,n_i+1}^i, E_{n_i,j}^i] - 2 \cdot 1_{i,t} e_{i,t} \\
&= -n_i e_{i,t} - 2 \cdot 1_{i,t} e_{i,t}.
\end{aligned}$$

We conclude that

$$\Omega_{i,t} = 1_{i,t}^2 + (n_i + 1) 1_{i,t}$$

satisfies $[\Omega_{i,t}, e_{i,t}] = -[\Omega_0, e_{i,t}]$. Thus, if we define

$$(8) \quad \Omega_1 = \Omega_0 + \sum_{i \in \mathbf{E}(D)} \Omega_{i,s} + \sum_{i \in \mathbf{E}(D)} \Omega_{i,t},$$

then $[\Omega_1, e_{i,s}] = [\Omega_1, e_{i,t}] = 0$ for all edges $i \in \mathbf{E}(D)$, and Ω_1 also commutes with each $\mathfrak{gl}_{n_i}$. Similar computations show Ω_1 commutes with $f_{i,s}$ and $f_{i,t}$ as well. It remains to determine the adjustment necessary to commute with e_v and f_v for vertices $v \in \mathbf{V}(D)$.

Again by Lemma 2.4 of [11],

$$\begin{aligned} \left[2 \sum_{\alpha \in \Delta'_+} \sum_j e_{-\alpha}^j e_{\alpha}^j, e_v \right] &= -2\alpha_v^\vee e_v \\ &= -2 \left(1'_v - \sum_{i \in S'_v} 1_{i,s} - \sum_{i \in T'_v} 1_{i,t} + (2 - N_v)1_v \right) e_v. \end{aligned}$$

For any edge $i \in S'_v$,

$$\begin{aligned} [\Omega_{i,s}, e_v] &= [1_{i,s}^2, e_v] + (n_i + 1)[1_{i,s}, e_v] \\ &= -1_{i,s}e_v - e_v 1_{i,s} - (n_i + 1)e_v \\ &= -2 \cdot 1_{i,s}e_v - (n_i + 2)e_v, \end{aligned}$$

and for any edge $i \notin S'_v$, we have $[\Omega_{i,s}, e_v] = 0$. Similar computations hold for $[\Omega_{i,t}, e_v]$, so

$$[\Omega_1, e_v] = -2 \cdot 1'_v e_v - 2(2 - N_v)1_v e_v - \sum_{i \in S'_v} (n_i + 2)e_v - \sum_{i \in T'_v} (n_i + 2)e_v.$$

As $1'_v$ is central,

$$[2 \cdot 1'_v 1_v, e_v] = 2 \cdot 1'_v [1_v, e_v] = 2 \cdot 1'_v e_v.$$

Also,

$$\begin{aligned} [(2 - N_v)1_v^2, e_v] &= (2 - N_v)(1_v e_v + e_v 1_v) \\ &= (2 - N_v)(2 \cdot 1_v e_v - e_v) \\ &= 2(2 - N_v)1_v e_v - (2 - N_v)e_v. \end{aligned}$$

Thus, if we define the adjustment term

$$\Omega_v = 2 \cdot 1'_v 1_v + (2 - N_v)1_v^2 + \left(2 + \sum_{i \in S'_v} (n_i + 1) + \sum_{i \in T'_v} (n_i + 1) \right) 1_v,$$

then

$$\begin{aligned} [\Omega_v, e_v] &= 2 \cdot 1'_v e_v + 2(2 - N_v)1_v e_v - (2 - N_v)e_v + \left(2 + \sum_{i \in S'_v} (n_i + 1) + \sum_{i \in T'_v} (n_i + 1) \right) e_v \\ &= 2 \cdot 1'_v e_v + 2(2 - N_v)1_v e_v + \left(\sum_{i \in S'_v} (n_i + 2) + \sum_{i \in T'_v} (n_i + 2) \right) e_v \\ &= -[\Omega_1, e_v]. \end{aligned}$$

It is clear that Ω_v commutes with e_w for any vertex $w \neq v$, as well as $e_{i,s}$, $e_{i,t}$, and $\mathfrak{gl}_{n_i}$ for any edge $i \in \mathbf{E}(D)$, as 1_v acts as 0 on all of them. Thus, if we define

$$(9) \quad \Omega = \Omega_1 + \sum_{v \in \mathbf{V}(D)} \Omega_v,$$

then Ω commutes with e_v for all $v \in \mathbf{V}(D)$, as well as $e_{i,s}$, $e_{i,t}$, and $\mathfrak{gl}_{n_i}$ for all $i \in \mathbf{E}(D)$. Similar computations show Ω commutes with f_v , $f_{i,s}$ and $f_{i,t}$. Thus, Ω commutes with all Chevalley generators, and hence all of $\mathfrak{g}(D, \mathbf{N})$. Its final form is

$$(10) \quad \begin{aligned} \Omega = & \sum_{i \in \mathbf{E}(D)} \sum_{j=1}^{n_i} \sum_{k=1}^{n_i} E_{k,j}^i E_{j,k}^i + 2 \sum_{\alpha \in \Delta'_+} \sum_j e_{-\alpha}^j e_{\alpha}^j \\ & + \sum_{i \in \mathbf{E}(D)} (1_{i,s}^2 + (n_i + 1)1_{i,s} + 1_{i,t}^2 + (n_i + 1)1_{i,t}) \\ & + \sum_{v \in \mathbf{V}(D)} \left(2 \cdot 1'_v 1_v + (2 - N_v)1_v^2 + \left(2 + \sum_{i \in S'_v} (n_i + 1) + \sum_{i \in T'_v} (n_i + 1) \right) 1_v \right). \end{aligned}$$

We now interpolate this construction to the Deligne category setting. Suppose we have a categorical Kac-Moody Lie algebra $\text{KM} = \text{KM}(D, \delta)$, which has a $\Gamma(D)$ -grading. We interpolate each term of Equation (10):

- For $i \in \mathbf{E}(D)$, the interpolation of $\sum_{j=1}^{n_i} \sum_{k=1}^{n_i} E_{k,j}^i E_{j,k}^i$ is the following composition:

$$\mathbb{1} \rightarrow \mathbb{1} \otimes \mathbb{1} \xrightarrow{\text{coev}_{V_i^*} \otimes \text{coev}_{V_i^*}} (V_i^* \otimes V_i) \otimes (V_i^* \otimes V_i) \xrightarrow{c_{(1234)}} (V_i \otimes V_i^*) \otimes (V \otimes V_i^*) \xrightarrow{m} \mathcal{U}_{\leq 2}(\text{KM}),$$

where m is the multiplication map.

- Instead of taking a sum of dual bases over roots α , we take this sum over positive degrees $\gamma \in \Gamma(D)$. Let $\Gamma_+(D) \subseteq \Gamma(D)$ denote the set of nonzero degrees with all coefficients nonnegative.

For $\gamma \in \Gamma_+(D)$, the pairing in Section 4.2 restricted to $\text{KM}_{-\gamma} \otimes \text{KM}_{\gamma}$ is a map

$$(\bullet, \bullet) : \text{KM}_{-\gamma} \otimes \text{KM}_{\gamma} \rightarrow \mathbb{1}.$$

As

$$\text{Hom}_{\mathcal{C}}(\text{KM}_{-\gamma} \otimes \text{KM}_{\gamma}, \mathbb{1}) \cong \text{Hom}_{\mathcal{C}}(\text{KM}_{-\gamma}, \text{KM}_{\gamma}^*),$$

it induces a map

$$(\bullet, \bullet)^{-1} : \text{KM}_{\gamma}^* \rightarrow \text{KM}_{-\gamma}.$$

Thus, to interpolate the sum over dual bases of $\text{KM}_{-\gamma}$ and KM_{γ} , we take the coevaluation map induced by the pairing, i.e., the composition

$$\mathbb{1} \xrightarrow{\text{coev}_{\text{KM}_{\gamma}^*}} \text{KM}_{\gamma}^* \otimes \text{KM}_{\gamma} \xrightarrow{(\bullet, \bullet)^{-1} \otimes \text{id}} \text{KM}_{-\gamma} \otimes \text{KM}_{\gamma} \xrightarrow{m} \mathcal{U}_{\leq 2}(\text{KM}).$$

The interpolation of $2 \sum_{\alpha \in \Delta'_+} \sum_j e_{-\alpha}^j e_{\alpha}^j$ is thus twice the sum of this map over all $\gamma \in \Gamma_+(D)$.

- The remaining terms all interpolate naturally. For example, the term $1_{i,s}^2$ interpolates to the map $\mathbb{1} \rightarrow \mathbb{1}_{i,s} \otimes \mathbb{1}_{i,s} \xrightarrow{m} \mathcal{U}_{\leq 2}(\text{KM})$, and the term $(n_i + 1)1_{i,s}$ interpolates to $n_i + 1$ times the map $\mathbb{1} \rightarrow \mathbb{1}_{i,s} \hookrightarrow \mathcal{U}_{\leq 2}(\text{KM})$.

The desired Casimir operator Ω can be written as the sum of the maps $\mathbb{1} \rightarrow \mathcal{U}_{\leq 2}(\text{KM})$ as above.

5. CATEGORY $\mathcal{O}$

Let $\text{KM} = \text{KM}(D, \delta)$ be an interpolated Kac-Moody Lie algebra with decomposition $\text{KM} = \mathfrak{u}_- \oplus \mathfrak{l} \oplus \mathfrak{u}_+$. Consider a KM-module M , i.e., an ind-object that KM acts on. A *weight* consists of the following data:

- for each edge $e \in \mathbf{E}(D)$, a bipartition (λ_e, μ_e) ,
- for each edge $e \in \mathbf{E}(D)$, scalars $c_{i,s}$ and $c_{i,t}$, and
- for each vertex $v \in \mathbf{V}(D)$, scalars c_v and c'_v .

For such a collection of bipartitions $(\underline{\lambda}, \underline{\mu})$ and scalars $\underline{c}$, we define the irreducible $\mathfrak{l}$ -module

$$L_{(\underline{\lambda}, \underline{\mu})}^{\underline{c}} = \bigotimes_{e \in \mathbf{E}(D)} L_{e, (\lambda_e, \mu_e)}$$

with $1_{i,s}$, $1_{i,t}$, 1_v , and $1'_v$ acting as scalars $c_{i,s}$, $c_{i,t}$, c_v , and c'_v , respectively. (Here, $L_{e, (\lambda_e, \mu_e)}$ is the irreducible object of $\text{Rep}(GL_{\mathbf{n}_{\delta e}})$ indexed by the bipartition (λ_e, μ_e) .) The *weight subobject* $M_{(\underline{\lambda}, \underline{\mu})}^{\underline{c}}$ of M with respect to this weight is the sum of the images of all morphisms of $\mathfrak{l}$ -modules $f \in \text{Hom}_{\mathfrak{l}}(L_{(\underline{\lambda}, \underline{\mu})}^{\underline{c}}, M)$, provided that the space of such morphisms is finite-dimensional. We may also refer to this subobject as M_{β} for the weight $\beta = (\underline{\lambda}, \underline{\mu}, \underline{c})$.

For such a weight β , we define its negative as $-\beta = (\underline{\mu}, \underline{\lambda}, -\underline{c})$.

The weights of KM as a module over $\mathfrak{l}$ (with nonzero weight subobjects) are *roots*. For a root $\alpha = (\underline{\lambda}, \underline{\mu}, \underline{c})$, we abbreviate the corresponding weight subobject of KM as KM_{α} .

For KM^* -modules, we define $(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})'$ similarly as an $\mathfrak{l}^*$ -module. Note that $(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})'$ and $L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}}$ are isomorphic objects.

Definition 5.1. *For the Deligne category setting, define the *parabolic category* $\mathcal{O}$ to be the category of finitely-generated $\text{KM}(D, \delta)$ -modules with a locally nilpotent $\mathfrak{u}_+$ -action and a weight decomposition, i.e., a decomposition of M into a direct sum of weight subobjects. Likewise, we define $\mathcal{O}^{\vee}$ the same way, except with a locally nilpotent $\mathfrak{u}_-$ -action instead.*

5.1. Verma Modules and the Action of the Casimir Operator. Recall that for a Kac-Moody algebra $\mathfrak{g}(A)$ and $\lambda \in \mathfrak{h}^*$, the *Verma module* $M(\lambda)$ is the highest weight module $\mathcal{U}(\mathfrak{g}(A))/J(\lambda)$, where $J(\lambda)$ is the left ideal generated by e_i for $i = 1, \dots, n$ and $h - \lambda(h)$ for all $h \in \mathfrak{h}$. It is easy to see that this is the module $\mathcal{U}(\mathfrak{g}(A)) \otimes_{\mathcal{U}(\mathfrak{n}_+ \oplus \mathfrak{h})} \mathbb{C}_{\lambda}$, where $\mathbb{C}_{\lambda}$ is the $\mathfrak{n}_+ \oplus \mathfrak{h}$ -module where $\mathfrak{n}_+$ acts by 0 and h acts by $\lambda(h)$ for all $h \in \mathfrak{h}$.

Analogously, for a parabolic subalgebra $\mathfrak{p}$ of $\mathfrak{g}(A)$ corresponding to a subdiagram I and highest weight irreducible representation V of $\mathfrak{p}$, the *parabolic Verma module* is $M_{\mathfrak{p}}(V) = \mathcal{U}(\mathfrak{g}(A)) \otimes_{\mathcal{U}(\mathfrak{p})} V$, where $\mathfrak{g}(A)$ acts on $\mathcal{U}(\mathfrak{g}(A))$ on the left.

Note that we also have the analogous notion of lowest-weight Verma modules, denoted $M^{\vee}(\lambda)$, which in the classical case is given by $\mathcal{U}(\mathfrak{g}(A))/J(\lambda)$, where $J(\lambda)$ is the left ideal generated by f_i for $i = 1, \dots, n$ and $h - \lambda(h)$ for all $h \in \mathfrak{h}$. The analogous lowest weight parabolic Verma module is denoted $M_{\mathfrak{p}}^{\vee}(V)$.

Let $M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}) = \mathcal{U}(\mathfrak{g}(A)) \otimes_{\mathcal{U}(\mathfrak{p})} L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}$ be the Verma module generated by the irreducible highest-weight object $L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}$. Since Ω commutes with all generators of $\text{KM}(D, \delta)$, its action on $M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$ is scalar-valued and equal to its action on $L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}$. Let $\Delta_I \subseteq \Delta$ denote the subset of roots generated by the subdiagram I , and $\Delta_{+, I} = \Delta_+ \cap \Delta_I$ denote the set of positive such roots. We denote the scalar actions of the degree-0 elements on $L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}$ as follows:

- (1) Positive Roots: For every $\alpha \in \Delta_+ \setminus \Delta_{+, I}$, $e_{\alpha}^{(k)} \cdot L_{\lambda, \mu} = 0$.
- (2) $\mathfrak{gl}_{n_i}$ Quadratic Element: By Schur's Lemma for each edge $i \in \mathbf{E}(D)$,

$$\left(\sum_{j,k=1}^{n_i} E_{k,j}^i E_{j,k}^i \right) \cdot L_{\lambda_i, \mu_i} = C_i L_{\lambda_i, \mu_i} \quad (C_i \in \mathbb{C}).$$

To compute the scalar C_i by which the $\mathfrak{gl}_{n_i}$ Casimir operator

$$\Omega_0^{\mathfrak{gl}_{n_i}} = \sum_{j,m=1}^{n_i} E_{m,j}^i E_{j,m}^i$$

acts on L_{λ_i, μ_i} , we interpolate the classical formula. For a classical Lie algebra $\mathfrak{gl}_{n_i}$, the corresponding highest weight is given by concatenating the partitions and filling the middle with zeros:

$$\gamma = (\lambda_{i,1}, \dots, \lambda_{i,k}, 0, \dots, 0, -\mu_{i,\ell}, \dots, -\mu_{i,1}).$$

On the classical irreducible highest weight module $V(\gamma)$, the Casimir operator acts by the scalar

$$\Omega_0^{\mathfrak{gl}_{n_i}}|_{V(\gamma)} = \sum_{j=1}^{n_i} \gamma_j(\gamma_j + n_i - 2j + 1) \text{Id}.$$

We split this sum into the nonzero components corresponding to λ (the first k terms) and μ (the last ℓ terms). For the first k terms, the contribution is exactly $\sum_{j=1}^k \lambda_{i,j}(\lambda_{i,j} + n_i - 2j + 1)$.

For the μ terms at the end of the weight vector, the index from the right is $n_i - j + 1$ (for $1 \leq j \leq \ell$), and the corresponding weight is $-\mu_{i,j}$. Substituting these into the summand gives:

$$(-\mu_{i,j})(-\mu_{i,j} + n_i - 2(n_i - j + 1) + 1) = \mu_{i,j}(\mu_{i,j} + n_i - 2j + 1).$$

Summing both contributions and factoring out the n_i terms yields:

$$\begin{aligned} C_i &= \sum_{j=1}^k \lambda_{i,j}(\lambda_{i,j} + n_i - 2j + 1) + \sum_{j=1}^{\ell} \mu_{i,j}(\mu_{i,j} + n_i - 2j + 1) \\ &= n_i \left(\sum_{j=1}^k \lambda_{i,j} + \sum_{j=1}^{\ell} \mu_{i,j} \right) + \sum_{j=1}^k \lambda_{i,j}(\lambda_{i,j} - 2j + 1) + \sum_{j=1}^{\ell} \mu_{i,j}(\mu_{i,j} - 2j + 1). \end{aligned}$$

(3) Edge Unit Objects: For each edge $i \in \mathbf{E}(D)$,

$$1_{i,s} \cdot L_{\underline{\lambda}, \underline{\mu}}^c = c_{i,s} L_{\underline{\lambda}, \underline{\mu}}^c, \quad 1_{i,t} \cdot L_{\underline{\lambda}, \underline{\mu}}^c = c_{i,t} L_{\underline{\lambda}, \underline{\mu}}^c \quad (c_{i,s}, c_{i,t} \in \mathbb{C}).$$

(4) Vertex Unit Objects: For each vertex $v \in V(D)$,

$$1_v \cdot L_{\underline{\lambda}, \underline{\mu}}^c = c_v L_{\underline{\lambda}, \underline{\mu}}^c, \quad 1'_v \cdot L_{\underline{\lambda}, \underline{\mu}}^c = c'_v L_{\underline{\lambda}, \underline{\mu}}^c \quad (c_v, c'_v \in \mathbb{C}).$$

Evaluating on each component of $L_{\underline{\lambda}, \underline{\mu}}^c$:

$$\begin{aligned} \Omega_0|_{L_{\underline{\lambda}, \underline{\mu}}^c} &= \sum_{i \in \mathbf{E}(D)} C_i, \\ \Omega_{i,s}|_{L_{\underline{\lambda}, \underline{\mu}}^c} &= c_{i,s}^2 + (n_i + 1)c_{i,s}, \\ \Omega_{i,t}|_{L_{\underline{\lambda}, \underline{\mu}}^c} &= c_{i,t}^2 + (n_i + 1)c_{i,t}, \\ \Omega_v|_{L_{\underline{\lambda}, \underline{\mu}}^c} &= 2c'_v c_v + (2 - N_v)c_v^2 + \left(2 + \sum_{i \in S'_v} (n_i + 1) + \sum_{i \in T'_v} (n_i + 1) \right) c_v. \end{aligned}$$

Adding together the scalar actions above gives us the scalar that Ω acts as. We then interpolate by replacing n_i with δ_i .

Proposition 5.2. *The Casimir operator Ω acts on the Verma module $M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$ by multiplication by the scalar*

$$\begin{aligned} \Omega|_{M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})} &= \sum_{i \in \mathbf{E}(D)} C_i + \sum_{i \in \mathbf{E}(D)} (c_{i,s}^2 + (\delta_i + 1)c_{i,s} + c_{i,t}^2 + (\delta_i + 1)c_{i,t}) \\ &\quad + \sum_{v \in V(D)} \left[2c'_v c_v + (2 - N_v)c_v^2 + \left(2 + \sum_{i \in S'_v} (\delta_i + 1) + \sum_{i \in T'_v} (\delta_i + 1) \right) c_v \right]. \end{aligned}$$

Proof. We established above that this is the scalar which Ω acts as on the generating part of $M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$, and Ω commutes with the action of $\text{KM}(D, \delta)$. $\square$

5.2. Shapovalov's Bilinear Form. For a Kac-Moody algebra $\mathfrak{g}(A)$ and Verma module $M(\lambda)$, there exists a unique up to scaling bilinear form $\langle \bullet, \bullet \rangle : M(\lambda) \times M(\lambda) \rightarrow \mathbb{C}$ which is contravariant, i.e. for any $g \in \mathfrak{g}(A)$ and $x, y \in M(\lambda)$, the form satisfies $\langle g \cdot x, y \rangle = -\langle x, \omega(g) \cdot y \rangle$. This form is called the *Shapovalov form*. We will generalize this form and recover its properties in the Deligne category setting.

Lemma 5.3. *If L is a Lie algebra object in a tensor category and V is an L -module, then V^* is naturally an L -module as well.*

Proof. From the L -action we have a morphism $\text{act} : L \otimes V \rightarrow V$. From here, it is not hard to see that we get an action $L \otimes V^* \rightarrow V^*$ via

$$L \otimes V^* \xrightarrow{\text{Id} \otimes \text{coev} \otimes \text{Id}} L \otimes V^* \otimes V \otimes V^* \xrightarrow{c_{L, V^*}} V^* \otimes L \otimes V \otimes V^* \xrightarrow{-\text{Id} \otimes \text{act} \otimes \text{Id}} V^* \otimes V \otimes V^* \xrightarrow{\text{ev} \otimes \text{Id}} V^*.$$

That this construction satisfies the Lie module axiom

$$\text{act}_{V^*} \circ ([\bullet, \bullet] \otimes \text{Id}_{V^*}) = \text{act}_{V^*} \circ (\text{Id}_L \otimes \text{act}_{V^*}) - \text{act}_{V^*} \circ (\text{Id}_L \otimes \text{act}_{V^*}) \circ (c_{L, L} \otimes \text{Id}_{V^*})$$

follows directly by dualizing the module axiom for V via the duality identity

$$\text{ev} \circ (\text{act}_{V^*} \otimes \text{Id}_V) = -\text{ev} \circ (\text{Id}_{V^*} \otimes \text{act}_V) \circ (c_{L, V^*} \otimes \text{Id}_V)$$

combined with the naturality of the braiding. $\square$

As before, let $F : \boxtimes \text{Rep}(GL_{\mathbf{n}_e}) \rightarrow \boxtimes \text{Rep}(GL_{\mathbf{n}_e})$ be the dualizing functor, and let KM^* be the functorial image of KM . Note that $M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$ has the structure of a KM^* -module obtained by precomposing the KM -action with $\omega^{-1} : (\text{KM})^* \rightarrow \text{KM}$. Through the grading by the Levi component, we have a decomposition

$$M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}) = \bigoplus_{\underline{\lambda}', \underline{\mu}', \underline{c}'} (L_{\underline{\lambda}', \underline{\mu}'}^{\underline{c}'})^{\oplus r(\underline{\lambda}', \underline{\mu}', \underline{c}')}.$$

Define the *restricted dual* of this module to be

$$M^*(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}) = \bigoplus_{\underline{\lambda}', \underline{\mu}', \underline{c}'} ((L_{\underline{\lambda}', \underline{\mu}'}^{\underline{c}'})^*)^{\oplus r(\underline{\lambda}', \underline{\mu}', \underline{c}')}.$$

Note that $(L_{\underline{\lambda}', \underline{\mu}'}^{\underline{c}'})^* \cong L_{\underline{\mu}', \underline{\lambda}'}^{-\underline{c}'}$ as $\mathfrak{l}$ -modules.

Proposition 5.4. *$M^*(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$ has a natural KM -module structure.*

Proof. For any root $\alpha = (\underline{\lambda}', \underline{\mu}', \underline{c}')$ and weight $\beta = (\underline{\nu}, \underline{\eta}, \underline{d})$, we have an induced action

$$\text{KM}_\alpha \otimes M^*(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})_\beta \rightarrow \bigoplus_{\beta' = (\underline{\nu}', \underline{\eta}', \underline{d}')} M^*(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})_{\beta'}.$$

The weights β' that appear in the sum on the right are those for which $\underline{d}' = \underline{c} + \underline{d}$ and both Littlewood-Richardson coefficients $c_{\lambda'_e, \nu_e}^{\nu'_e}$ and $c_{\mu'_e, \eta_e}^{\eta'_e}$ are nonzero for all edges $e \in \mathbf{E}(D)$. We denote this as $\beta' \sim \alpha \blacktriangle \beta$.

Interpolating from the standard Kac-Moody setting, we can find a map ρ^* such that the following diagram commutes:

$$\begin{array}{ccc}
\mathrm{KM}_\alpha \otimes M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})^*_{\beta} \otimes \bigoplus_{\beta' \sim (-\alpha) \blacktriangle \beta} M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})_{\beta'} & \xrightarrow{c_{(12)}} & M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})^*_{\beta} \otimes \mathrm{KM}_\alpha \otimes \bigoplus_{\beta' \sim (-\alpha) \blacktriangle \beta} M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})_{\beta'} \\
\downarrow \rho^* \otimes \mathrm{Id} & & \downarrow -\mathrm{Id} \otimes \mathrm{act} \\
\left(\bigoplus_{\beta' \sim (-\alpha) \blacktriangle \beta} M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})^*_{\beta'} \right) \otimes \left(\bigoplus_{\beta' \sim (-\alpha) \blacktriangle \beta} M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})_{\beta'} \right) & & M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})^*_{\beta} \otimes M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}) \\
\downarrow \mathrm{ev} & & \downarrow \pi \\
\mathbb{1} \xleftarrow{\mathrm{ev}} M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})^*_{\beta} \otimes M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})_{\beta} & &
\end{array}$$

where π is the corresponding projection. This proves that the restricted dual has the structure of a KM-module. $\square$

Proposition 5.5. *We have the following isomorphisms:*

- (1) $\varphi : M^\vee(L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}}) \rightarrow M^*(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$ as an isomorphism of KM-modules.
- (2) $\psi : M((L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})') \rightarrow M^\vee(L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}})$ as an intertwining isomorphism between the action of KM^* and the action of KM.

Proof.

- (1) The lowest-weight Verma module $M^\vee(L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}})$ is generated by its lowest-weight space $L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}}$, which KM_{-1} annihilates.

The highest-weight Verma module $M(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$ is generated by its highest-weight object $L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}$, which KM_1 annihilates. When we take the restricted dual $M^*(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$, the grading of the module is inverted. The highest-weight object $L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}$ dualizes to the lowest-weight object $(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})^* \cong L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}}$. Since KM_1 annihilates $L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}}$, KM_{-1} annihilates this lowest-weight object. Thus, the universal property of the lowest-weight Verma module guarantees the existence of a canonical KM-module homomorphism

$$\varphi : M^\vee(L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}}) \rightarrow M^*(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$$

that uniquely extends the identity map on the lowest-weight object $L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}}$. As this lowest-weight object generates $M^*(L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})$, φ is surjective.

For sufficiently large integer parameters δ_e , φ is an isomorphism due to the comparison of the corresponding $\Gamma(D)$ -graded parts of both characters (see e.g. [10, 14]). Therefore, it is an isomorphism for all δ_e .

- (2) We will construct this isomorphism by constructing it separately at each weight. For each weight $\beta = (\underline{\lambda}', \underline{\mu}', \underline{c}')$ of $M((L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})')$, we want an isomorphism

$$\psi_\beta : M((L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})')_\beta \rightarrow M^\vee(L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}})_{-\beta}$$

between weight subobjects such that for all roots $\alpha = (\underline{\lambda}'', \underline{\mu}'', \underline{c}'')$, the following diagram commutes:

$$\begin{array}{ccccc} \mathrm{KM}_\alpha^* \otimes (L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})' & \xrightarrow{\mathrm{act}^*} & M((L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})') & \xrightarrow{\pi'} & M((L_{\underline{\lambda}, \underline{\mu}}^{\underline{c}})'_\beta) \\ \downarrow \omega^{-1} \otimes \mathrm{Id} & & & & \downarrow \psi_\beta \\ \mathrm{KM}_{-\alpha} \otimes L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}} & \xrightarrow{\mathrm{act}} & M^\vee(L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}}) & \xrightarrow{\pi} & M^\vee(L_{\underline{\mu}, \underline{\lambda}}^{-\underline{c}})_{-\beta} \end{array}$$

where π and π' denote the respective projections. It is only necessary to check this for roots α such that $\beta \sim \alpha \blacktriangleleft (\underline{\lambda}, \underline{\mu}, \underline{c})$. In particular, this only needs to be verified for finitely many roots α .

Fixing β , for sufficiently large integer parameters δ_e , the isomorphism ψ_β exists (as the two weight subobjects have equivalent descriptions by PBW). Thus, it exists for all δ_e . $\square$

Thus, we define the unique up to rescaling contravariant *Shapovalov form* to be the pairing between the KM^* -module and the KM -module given by

$$(11) \quad \mathcal{S} : M(L_{\lambda, \mu}) \otimes M(L_{\lambda, \mu}) \xrightarrow{\psi \otimes \mathrm{Id}} M^\vee(L_{\mu, \lambda}) \otimes M(L_{\lambda, \mu}) \xrightarrow{\varphi \otimes \mathrm{Id}} M^*(L_{\lambda, \mu}) \otimes M(L_{\lambda, \mu}) \xrightarrow{\mathrm{ev}} \mathbb{1}.$$

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NOMENCLATURE

- $(,)$ Invariant bilinear form (Theorem 2.1 and Section 4.2)
- $1_{i,s}; 1_{i,t}; 1_v; 1'_v$ Edge and vertex Cartan Chevalley generators (Section 3.2 and equation (5))
- $\mathbb{1}_{i,s}; \mathbb{1}_{i,t}; \mathbb{1}_v; \mathbb{1}'_v$ Cartan Levi generators (Definition 3.8)
- $\mathrm{ev}_X; \mathrm{coev}_X; c_{X,Y}$ Evaluation, coevaluation morphisms and symmetric braiding (Section 2.2)
- $\mathfrak{g}(D, \mathbf{N})$ Kac-Moody Lie Algebra (Section 3.1)
- $\Gamma(D)$ Free abelian group generated by $\mathbf{V}'(D)$ (Section 3.1)
- $\mathrm{KM}(D, \delta)$ Kac-Moody Lie algebra in Deligne category (Definition 3.8)
- $\mathbf{V}(D); \mathbf{V}'(D); \mathbf{E}(D); \mathbf{L}(D)$ Vertices (added), edges and self-loops of the graph D (Section 3.1)
- $\mathcal{O}$ Parabolic category $\mathcal{O}$ (Definition 5.1)
- $\mathcal{S}$ Shapovalov form (equation (11))
- Ω Casimir element in $\mathfrak{g}(D, N)$ (Equation (10))
- ω Cartan involution (equation (1) and Section 4.1)
- $\mathrm{Rep}(GL_t)$ Deligne category (Section 2.3)
- $e_{i,s}; f_{i,s}; e_{i,t}; f_{i,t}; e_v; f_v$ Edge and vertex Chevalley generators (Section 3.1)
- $E_{j,k}^i$ Matrix element of an edge i (Section 3.1)
- F Duality autoequivalence functor (Section 4.1)
- FL Free Lie algebra (Section 2.2 and equation (2))
- $M(\lambda); M_{\mathbf{p}}(V); M(L_{\lambda, \mu}); M^\vee(L_{\lambda, \mu}); M^*(L_{\lambda, \mu})$ (parabolic) Verma (w.r.t. $\mathbf{u}_\pm$) modules and the restricted duals (Section 5.1)
- $S_v; T_v; L_v; S'_v; T'_v$ Source, target, loop edges, incoming/outgoing arrows of a vertex v (Section 3.1)
- $V_{i,s}; V_{i,s}^*; V_{i,t}; V_{i,t}^*; \mathbb{1}_{v,\pm}$ Generating Chevalley components of the Kac-Moody Lie algebra (Proposition 3.1)

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